



**MEENAKSHI SUNDARARAJAN
ENGINEERING COLLEGE**

AFFILIATED TO ANNA UNIVERSITY | AN AUTONOMOUS INSTITUTION



TNEA CODE 1309

PROCEEDINGS OF THIRD RESEARCH SUMMIT 2025

**INNOVATE, SIMULATE & SOLVE
(RSISS'25)**

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**SRI SHANKARA
MEENAKSHI SUNDARARAJAN
RESEARCH CENTRE**

A RECOGNIZED RESEARCH CENTER BY ANNA UNIVERSITY



Proceedings of Third Research Summit 2025
Innovate, Simulate & Solve (RSISS'25)



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Editors

Dr. Saravanan S.V

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Dr. Arivazhagan R

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Published by : KIM Scientific Publications, Chennai, India
Meenakshi Sundararajan Engineering College (Autonomous)
Arcot Road, Kodambakkam, Chennai - 600 024, Tamil Nadu, India
www.msec.edu.in

ISBN: 978-93-94412-32-3

Meenakshi Sundararajan Engineering College (Autonomous), Chennai - 600 024
ISBN: 978-93-94412-32-3



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About the Principal



Dr. Saravanan S.V is the Principal and Senior Professor with over 32 years of teaching, research, and administrative experience. He holds B.E., M.E., Ph.D. in Mechanical Engineering, M.B.A., and an Honorary D.Litt. degree, has guided seven Ph.D. scholars, mentored more than 100 university rank holders, holds published patent, and has authored textbooks with over 100 research publications. He serves as an editorial board member, has organized and chaired several national and international programs and conferences, and has received numerous prestigious academic and leadership awards.

About the Editors



Dr. Poongothai S is the Dean – Research & Welfare & Wellness (Learning Community) & former Professor & Head of Civil Engineering, Annamalai University with over 35 years of academic and research experience. She holds a Ph.D. from Anna University under QIP and M.E. degrees in Structures and Water Resources Engineering & Management, with extensive research output including International & National Journal & Conference publications more than 90, and a UGC-MRP funded project on groundwater quality modeling Co-Coordinator of One Crore UGC-SAP Project. She has guided 11 Ph.D. scholars and PG thesis, delivered numerous invited lectures, reviewer of reputed Journals and actively contributes to academic leadership, professional bodies, and research mentorship.



Dr. Arulkarthick V. J is Dean of the Internal Quality Assurance Cell and Professor in the Department of Electronics and Communication Engineering. He holds Ph.D. with a demonstrated history of working in the higher education industry. His research focuses on VLSI design, Image Processing and Communication Systems. His interest also includes Educational research, E-Learning, Faculty Development, Educational Leadership, and Classroom Management, Quality Assurance, Ranking & accreditation.



Dr. Arivazhagan R is Professor and Head of Research in the Department of Civil Engineering. He has 35 years of total experience, bringing extensive knowledge and expertise to academics and research. He holds an A.M.I.E. from The Institution of Engineers (India), an M.Tech in Coastal Management with University Second Rank from Anna University, and a Ph.D. in Geotechnical Engineering. He has made significant contributions to the field of Civil Engineering through teaching and research. He continues to guide and mentor future engineers with unwavering dedication.



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Mr. Mohan Raj Vijayan is an Assistant Professor in the Department of Information Technology. He holds an M.E. in Computer Science and Engineering and is currently pursuing a Ph.D. in Knowledge-Based Systems at DIST, CEG, Anna University. His multidisciplinary qualifications span Management, Music, Psychology, Philosophy, French, and Archaeology & Epigraphy. With nearly two decades of experience in academia, research, and industry, he brings strong practical and theoretical expertise, and his research interests include Generative AI, Intelligent Agents, Knowledge Engineering, Computational Linguistics, and Indian Knowledge Systems.



Dr. Sivasubramaniyan S.B is currently serving as an Assistant Professor in the Department of Electrical and Electronics Engineering. With 13.5 years of total experience, he has made significant contributions to the field. His educational journey includes an undergraduate degree in Electrical and Electronics Engineering from the University of Madras, where he graduated with First Class in 2002. He further obtained an M.E. in VLSI Design from Anna University, also achieving First Class in 2009. His dedication to research culminated in a PhD in the faculty of Information and Communication Engineering, which was awarded in April 2023.

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About the Technical Editors



Dr. Sowmya S holds a Ph.D. in Biomedical Signal Processing from Anna University, with research focused on wearable devices and deep learning models for maternal health monitoring having over 17 years of academic experience. Her contributions include publications in reputed SCI and Scopus-indexed journals, patents in healthcare technology, and active involvement as a reviewer for leading international journals. She is passionate about applying AI for real-time health diagnostics and has received awards for both academic excellence and innovative research.



Dr. Sandhya C is a faculty member in the Department of Electronics and Communication Engineering with a strong interdisciplinary research focus. Her research spans bio signal and medical image processing, cognitive neuroscience, machine learning, and neuropsychology, with particular emphasis on speech and mental imagery for brain-computer interface applications. She is dedicated to bridging engineering and cognitive neuroscience through research, teaching, and mentoring, contributing to advancements with clinical and technological impact.



Mrs. Geetha G is a multidisciplinary professional with strong experience in both DevOps engineering and computer science academics. She holds an M.E. in Computer Science and Engineering from Anna University with a high academic rank. She is committed to continuous learning, research, system reliability, and delivering high-quality technical solutions.



Dr. Saji K.S holds a B.E. in EEE from Manonmaniam Sundaranar University, an M.E. in Control and Instrumentation Engineering from Annamalai University, and a PhD in Electrical Engineering from Anna University. She is a Professor in the EEE Department at Meenakshi Sundararajan Engineering College, Chennai., with over 24 years of teaching and research experience. Her research interests include Electrical Engineering, Controller Design, Electric Vehicle Technology, and Intelligent Control Algorithms. Under her supervision, one research scholar has completed their PhD, two have submitted their thesis and Six are currently pursuing their PhD at Anna University.



VISION AND MISSION OF THE INSTITUTE

VISION

To impart state-of-the art technical education, including sterling values and shining character, producing engineers who contribute to nation building thereby achieving our ultimate objective of sustained development of an unparalleled society, nation and world at large.

MISSION

Meenakshi Sundararajan Engineering College, Chennai constantly strives to be a Centre of Excellence with the singular aim of producing students of outstanding academic excellence and sterling character to benefit the society, our nation and the world at large.

To achieve this, the college ensures

- Continuous upgradation of its teaching faculty to ensure a high standard of quality education and to meet the ever-changing needs of the society.
- Constant interaction with its stakeholders.
- Linkage with other educational institutions and industries at the national and international level for mutual benefit.
- Provision of research facilities and infrastructure in line with global trends.

Adequate opportunities and exposure to the students through suitable programs, to mould their character and to develop their personality with an emphasis on professional ethics and moral values.



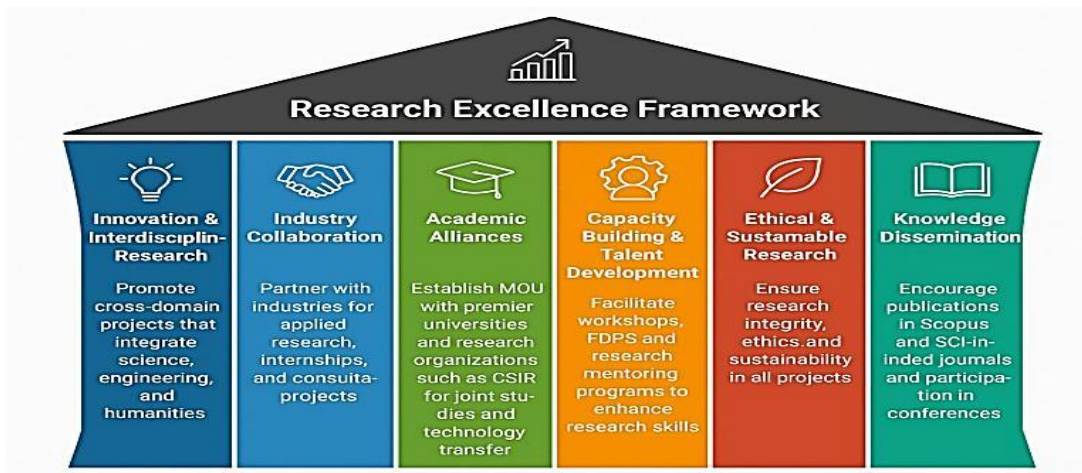
VISION AND MISSION OF THE RESEARCH CENTRE

VISION

To foster a vibrant research culture that promotes innovation, collaboration, and societal impact through interdisciplinary, industry-linked, and globally recognized research initiatives.

MISSION

- To foster a vibrant ecosystem of research excellence that empowers students and faculty in intellectual exploration and discovery.
- To advance knowledge and drive innovative solutions through impactful, publication-oriented research.
- To establish strong collaborations with national and international organizations, enhancing interdisciplinary engagement.
- To align institutional research initiatives with global Sustainable Development Goals (SDGs) and emerging technologies.
- To elevate the institution's research visibility, quality, and ranking through high-impact publications and citations.
- To contribute meaningfully to academia, industry, and the overall advancement of humankind.



RESEARCH PILLARS



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23rd December 2025

**Meenakshi Sundararajan Engineering College (Autonomous), Chennai,
Tamil Nadu – 600 024.**

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www.msec.edu.in

Meenakshi Sundararajan Engineering College (Autonomous), Chennai - 600 024
ISBN: 978-93-94412-32-3



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Book Title	: Proceedings of Third Research Summit 2025 Innovate, Simulate & Solve (RSISS'25)
Editor(s) Name	: Dr. Saravanan S.V, Principal & Senior Professor Dr. Poongothai S, Dean- Research Dr. Arul Karthick V. J, Dean - IQAC Dr. Arivazhagan R, Head - Research Mr. Mohan Raj Vijayan, Asst. Prof, IT Dr. Sivasubramaniyan S.B, Asst. Prof, EEE
Published By	: KIM Scientific Publications, Chennai, India
Printer's Address	: Meenakshi Sundararajan Engineering College (Autonomous) Chennai – 600 024, Tamil Nadu, India
Edition Details	: AI&DS/ Civil/ CSE/ ECE/ EEE/ IT/ MECH CHE/ENG/PHY/MATHS/TAMIL
Month & Year	: December 2025
ISBN	 9 789394 412323
Copyright 	: Meenakshi Sundararajan Engineering College
Website	www.msec.edu.in

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A Source of Inspiration and Support



Our Beloved Secretary

Mr. N. Sreekanth B.E., M.S.

Meenakshi Sundararajan Engineering College, (Autonomous)



Proceedings of Third Research Summit 2025 **Innovate, Simulate & Solve (RSISS'25)**



Dr. Saravanan S.V
Principal & Senior Professor

Message

“The best research will give the best teaching, and the best teaching will give the best research”

It gives me great pleasure to welcome you all to the Third Research Summit 2025 of our engineering college. This summit serves as a prestigious platform where innovative ideas, emerging technologies, and visionary research efforts come together.

In an era driven by knowledge, discovery, and rapid technological advancement, research is the cornerstone that shapes the future of engineering education. Our institution has consistently encouraged students and faculty to pursue research that is both socially relevant and academically rigorous. This summit reflects our continued commitment to nurturing a vibrant research culture on campus.

I extend my appreciation to all the presenters, participants, faculty mentors, and Dean- Research and organisers whose dedication has made this event possible and the proceedings with the ISBN. I am confident that the interactions, discussions, and collaborations emerging from this summit will inspire new perspectives and contribute meaningfully to the global scientific community.

Wishing you all a productive and enriching experience.

Meenakshi Sundararajan Engineering College (Autonomous), Chennai - 600 024
ISBN: 978-93-94412-32-3



Proceedings of Third Research Summit 2025

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Dr. Poongothai S
Dean -Research

Message

It gives me immense pleasure to extend my warm greetings to all participants of the Third Research Summit – RSISS '25, hosted by the Sri Shankara Meenakshi Sundararajan Research Centre (SSMSRC) at Meenakshi Sundararajan Engineering College. This summit stands as a distinguished platform celebrating research excellence, collaborative innovation, and transformative engineering solutions.

The theme of this year is “Innovate, Simulate & Solve” captures the essence of modern research and the spirit of technological advancement shaping the world today. Innovation fuels new ideas, simulation empowers experimentation and validation, and solution-driven engineering transforms concepts into impactful realities. Together, these principles form the foundation of progressive research that addresses societal and industrial challenges. This summit brings together brilliant minds researchers, academicians, industry experts, and students to exchange knowledge, explore emerging technologies, and engage in purposeful discussions.

I sincerely appreciate the relentless efforts of the authors, reviewers, session chairs, coordinators, and the organizing committee whose dedication has shaped this remarkable academic gathering. My heartfelt congratulations to all contributors for enriching our research culture with high-quality scholarly work. Let this summit be an inspiration to think beyond boundaries, simulate without constraints, and solve with purpose, advancing research that benefits society and humanity. I wish RSISS'25 grand success and encourage every participant to seize this opportunity to collaborate, innovate, and lead the future of engineering.

I profusely thank our beloved Secretary ,Principal ,Heads of the department and Management for their consistent support to bring this Research Summit a phenomenal success.



Proceedings of Third Research Summit 2025

Innovate, Simulate & Solve (RSISS'25)



Dr. Ramajeyam L
Dean -PG Studies

Message

The Research Summit at Meenakshi Sundararajan Engineering College reflects our institution's enduring commitment to academic excellence, innovation, and societal progress. This year's collection of research contributions highlights the intellectual curiosity, technical expertise, and collaborative spirit that define our students and faculty. From emerging engineering technologies to impactful interdisciplinary studies, each project embodies our vision of fostering knowledge that addresses real-world challenges and promotes sustainable advancement.

At MSEC, we believe that research is not merely an academic endeavor but a catalyst for transformation. The works showcased in this summit illustrate the power of inquiry-driven learning, creative thinking, and persistent exploration. I extend my appreciation to all researchers, mentors, and organizers whose dedication has elevated the standard of scholarship within our college.

I invite every participant to engage deeply with the ideas presented, exchange perspectives, and forge new pathways for innovation. May this summit inspire continued excellence and strengthen our collective pursuit of engineering solutions that benefit society.



Proceedings of Third Research Summit 2025

Innovate, Simulate & Solve (RSISS'25)



Dr. Arul Karthick V.J
Dean -IQAC

Message

It gives me great pleasure to present the Proceedings of the Research Contributions submitted by scholars, faculty members, and students. This compilation reflects the institution's continuous commitment toward academic excellence, innovation, and quality enhancement.

The Internal Quality Assurance Cell (IQAC) consistently strives to promote a culture of research and to ensure that scholarly activities align with our vision of achieving national and global standards. These proceedings stand as a testament to the dedication, hard work, and intellectual strength of our academic community.

I appreciate the efforts of all contributors, coordinators, reviewers, and organizing members who have worked diligently to bring this volume to fruition. I wish all researchers continued success in their academic pursuits and encourage them to carry forward the spirit of inquiry and innovation.



Proceedings of Third Research Summit 2025

Innovate, Simulate & Solve (RSISS'25)



Dr. Arivazhagan R
Head - Research

Message

This Research Summit provides a premier multidisciplinary platform for researchers, scholars, educators, and students to present and discuss the latest innovations, emerging trends, and practical challenges in the fields of Engineering and Technology. It also highlights the solutions and advancements adopted to address these challenges. The Summit serves as a bridge connecting students with industry, exposing them to real-world problems, high-level innovations, advanced laboratories, and meaningful research opportunities that enable them to work on real-time solutions.

This Summit showcases the significant progress our institution has made in strengthening and enhancing our research and publication capabilities. It also inspires us to pursue impactful innovations that will contribute to the betterment of society and mankind. We would like to express our sincere gratitude to all the authors and reviewers whose contributions have enabled us to maintain the high quality of the manuscripts published in the SSMSRC proceedings.

We remain optimistic that the proceedings of Research Summit 2026 will, in due course, be indexed in IEEE Xplore, Web of Science, and Scopus. We appreciate that the authors of Research Summit 2025 may wish to enhance the visibility and reach of their work, and we assure them of our full support in these efforts. We trust that the findings and recommendations emerging from this Summit will serve as a strong impetus for further research and scholarly exploration in these areas. We extend our heartfelt thanks to all authors and participants for their valuable contributions.



Proceedings of Third Research Summit 2025

Innovate, Simulate & Solve (RSISS'25)



Prof. Raman S
Department of Mathematics

Message

It gives me immense happiness to see our third Research Summit emerge as a collective expression of curiosity, responsibility, and humaneness. For twenty-five years, our institution has grown on a foundation of values, and this summit reflects that heritage.

The research presented here is not merely academic each contribution seeks to address real challenges, offer sustainable solutions, and serve society with integrity. I deeply appreciate the dedication of our faculty, whose work continues to inspire our students and uphold the ethical spirit that defines our college. May this pursuit of knowledge guide us toward a brighter, more compassionate future.



Proceedings of Third Research Summit 2025 **Innovate, Simulate & Solve (RSISS'25)**



PREFACE

It is with great pleasure that we present the Proceedings of the Third Research Summit 2025 to be held on December 23, 2025 at Meenakshi Sundararajan Engineering College. This annual summit serves as a significant platform for promoting scholarly exchange, showcasing innovative research, and encouraging academic engagement among students, faculty members, and researchers.

The Third Research Summit reflects the institution's continued commitment to advancing research-driven learning and fostering a culture of inquiry across diverse engineering and interdisciplinary domains. The abstracts included in this volume have undergone a rigorous review process to ensure academic quality and relevance. The breadth of topics covered demonstrates the expanding scope of research activities carried out within the college and in collaboration with the wider academic community. This proceeding aims to document the intellectual contributions presented at the summit and to inspire future investigations, collaborations, and innovations. It is our hope that the research compiled here will contribute meaningfully to ongoing academic discourse and provide a foundation for continued advancements in engineering and technology.

The Editorial Committee extends its sincere appreciation to all contributors whose efforts have enriched the Proceedings of the Third Research Summit 2025 of Meenakshi Sundararajan Engineering College. We thank all authors for their scholarly submissions and the reviewers for their meticulous evaluations and constructive feedback, which have helped uphold the academic quality of this publication. We also acknowledge the invaluable support of the session chairs, faculty coordinators, technical staff, and student volunteers whose dedication has ensured the smooth conduct of the summit. Our heartfelt gratitude goes to the Management of Meenakshi Sundararajan Engineering College for their unwavering encouragement and for providing the necessary resources to facilitate this academic endeavour.

The Research Summit provides an opportunity to explore emerging ideas, share innovative findings, and engage in meaningful discussions across various domains of engineering and technology. We hope that today's deliberations will spark new perspectives, inspire collaborative endeavours, and contribute to the continued advancement of research within our institution.

We extend our best wishes to all participants for a productive and enriching experience.

-Editorial Committee

Third Research Summit 2025

Meenakshi Sundararajan Engineering College



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Department of Artificial Intelligence and Data Science



Mrs. Mathangi N
Head

Message

As Head of the Department, I am proud to witness our faculty and students consistently pushing the boundaries of knowledge through impactful research. Our department fosters a culture of curiosity, collaboration, and innovation that translates into meaningful real-world solutions.

Each research contribution reflects our commitment to academic excellence and societal advancement. We continue to encourage interdisciplinary exploration and nurture emerging scholars. Together, we aim to build a vibrant research ecosystem that inspires progress and discovery.



Proceedings of Third Research Summit 2025
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Department of Civil Engineering



Dr. Asha P
Head

Message

It gives me great pleasure to extend my warm greetings to all participants of this year's Research Summit. This summit stands as a platform where ideas converge, perspectives deepen, and innovative thinking takes flight. As we witness rapid advancements across disciplines, the role of research has never been more vital. It is through rigorous inquiry and collaborative exploration that we shape solutions for the challenges of tomorrow.

Our institution has always encouraged a culture of curiosity, critical thinking, and multidisciplinary engagement. I am proud of the dedication shown by our faculty, scholars, and students who continuously strive to push the boundaries of knowledge. Events like this reflect our shared commitment to academic excellence and meaningful societal impact. I encourage all participants to engage wholeheartedly—ask bold questions, share your insights generously, and forge collaborations that extend beyond this summit. Let this gathering inspire new ideas, strengthen research ecosystems, and open pathways for impactful innovation.

My sincere appreciation goes to the organizers, contributors, and distinguished speakers for making this summit possible. I wish everyone an enriching and inspiring experience.



Proceedings of Third Research Summit 2025
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Department of Computer Science and Engineering



Dr. Arthi S
Head

Message

It is my pleasure to extend warm greetings to all participants of the Research Summit 2025. This forum provides an excellent opportunity for sharing innovative ideas, discussing emerging trends, and encouraging collaborative research in Computer Science and Engineering.

Our department remains committed to fostering a strong research culture and empowering students and faculty to contribute meaningfully to the rapidly evolving technological landscape.

I appreciate the efforts of the organizing committee, reviewers, and contributors who have worked diligently to make this event impactful. I also congratulate all authors for their scholarly contributions. May this summit ignite new collaborations, inspire fresh ideas, and strengthen our collective commitment to academic excellence and societal advancement.

I wish the Research Summit great success and look forward to seeing the innovations and discussions that will emerge from this gathering.



Proceedings of Third Research Summit 2025
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Department of Electronics and Communication Engineering



Mrs. Siji Sivanandhan
Head

Message

The summit aims to bring together experts, researchers, and students to share their knowledge and ideas on cutting-edge technologies. The summit will feature paper presentations on topics including Artificial Intelligence, IoT, VLSI, and Renewable Energy. Selected papers will be published in our college proceedings. Looking forward to your participation.



Proceedings of Third Research Summit 2025
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Department of Electrical and Electronics Engineering



Dr. Maya R
Head

Message

It is my great pleasure to extend greetings to all attendees of the Research Summit 2026. This Summit will be an excellent opportunity, will contribute to the advancement of research, knowledge gathering, and collective effort in the Field of AI.

The Department of Electrical and Electronics Engineering is committed to the Technological Advancements, Impactful Innovations, Addressing the challenges in the Society. Artificial Intelligence has Impact on all sectors of society, this conference will contribute to its advancement and advanced research.

My heartfelt congratulations to the all the organizers, authors, editors and reviewers. I wish the research summit will achieve the great success. Wishing everyone enriching and enlightening experience.



Proceedings of Third Research Summit 2025
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Department of Information Technology



Dr. Kanmozhi A
Head

Message

It is my pleasure to extend warm greetings to all participants of the Research Summit 2026. This gathering offers a meaningful opportunity for the exchange of knowledge, exploration of industrial advancements, and the encouragement of collaborative endeavors in Information Technology.

Our department remains committed to cultivating a dynamic research culture and enabling both students and faculty to contribute significantly to the fast-changing technological landscape. I sincerely commend the dedication of the organizing team, reviewers, and all contributors whose collective efforts have made this event truly impactful. My heartfelt congratulations also go to the authors for their noteworthy academic contributions.

May the Research Summit 2026 achieve outstanding success, inspire innovative thinking, spark fruitful partnerships, and reinforce our shared pursuit of academic excellence and societal growth.



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Department of Mechanical Engineering



Dr. Santhanakrishnan S
Head

Message

It is my pleasure to extend warm greetings to all participants of the Research Summit 2025. This forum provides an excellent opportunity for sharing innovative ideas, discussing emerging trends, and encouraging collaborative research in Mechanical Engineering. Our department remains committed to fostering a strong research culture and empowering students and faculty to contribute meaningfully to the rapidly evolving technological landscape.

I appreciate the efforts of the organizing committee, reviewers, and contributors who have worked diligently to make this event impactful. I also congratulate all authors for their scholarly contributions. May this summit ignite new collaborations, inspire fresh ideas, and strengthen our collective commitment to academic excellence and societal advancement.

I wish the Research Summit great success and look forward to seeing the innovations and discussions that will emerge from this gathering.



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Department of Sciences and Humanities



Mrs Vanathi T
Head

Message

I am very much excited that our 'Sri Shankara Meenakshi Sundararajan Research Centre' is hosting the *3rd Research Summit* on 23 December 2025. I am sure this summit would offer an excellent platform to showcase our work, exchange ideas across disciplines, and strengthen our research community. Looking forward to a vibrant and productive summit. All the very best!



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MSEC-RS-2025-PR-001

An Analytical Study on the Work–Life Balance (WLB) of Women Employees in Self-Financing Engineering Colleges

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Abstract

Because it is essential to creating and sustaining a productive work environment, work-life balance, or WLB, has become more important in both public and private sector businesses. People find it difficult to perform well at work without juggling their personal obligations due to the growing demands of modern occupations. Research indicates that non-work activities have a favourable impact on employee productivity and creativity, and social engagement outside of the workplace is not only a formality but also a significant stress reliever. Policymakers, employers, and workers all work to increase employee involvement in more fruitful ways. Effective task planning and organization makes it possible for employees to meet their personal obligations while juggling work and personal obligations. But work-life balance is still a personal concept that differs from person to person. This study looks at the work-life balance of female employees at self-financed engineering colleges and identifies the main variables that have a big impact on how well they can balance work and family obligations. Additionally, the study suggests ways that businesses and workers might improve overall work-life balance and deal with work-family issues.

Keywords- *Work–Life Balance, Quality of Life, Personal Commitments.*



MSEC-RS-2025-DR-002

AI-Enabled Groundwater Quality Monitoring and Management

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Abstract

Groundwater quality is a critical determinant of public health and water security in rapidly developing regions. In districts such as Cuddalore, Tamil Nadu, nitrate and fluoride contamination varies significantly across pre-monsoon, monsoon, and post-monsoon periods, challenging conventional monitoring practices. Existing Water Quality Index (WQI) and standalone machine-learning models provide only static predictions and do not translate results into actionable mitigation strategies for decision makers. To address this limitation, this study proposes an AI-enabled groundwater monitoring and management framework that integrates seasonal hydrochemical datasets with advanced predictive modeling and intelligent decision support. Machine learning algorithms are employed to classify wells and forecast contamination levels, while a Large Language Model (LLM) automatically interprets outputs, generates block-level risk summaries, and recommends appropriate treatment and intervention strategies.

Keywords: *Groundwater Quality, Nitrate and Fluoride Contamination, Machine Learning, Large Language Models (LLMs), Agentic AI and Decision Support System*



MSEC-RS-2025-DP-003

Behaviour of Cold-Formed Steel–Concrete Composite Sections

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Abstract

Cold-formed steel–concrete composite sections have emerged as an efficient structural solution for modern construction due to their favourable strength-to-weight ratio, ease of fabrication, and improved structural performance under various loading conditions. This study investigates the mechanical behaviour, load-carrying capacity, and failure mechanisms of cold-formed steel members acting compositely with concrete. The interaction between steel and concrete components is examined, with emphasis on shear transfer mechanisms, local and global buckling phenomena, and the influence of connection detailing, including embossments, screws, and shear connectors. Experimental results are complemented by finite-element simulations to evaluate stiffness, ultimate strength, and ductility characteristics. The findings reveal that composite action significantly enhances flexural strength and delays local buckling compared to bare cold-formed steel sections. Furthermore, the confinement provided by the steel improves the cracking behaviour and compressive response of the concrete core. The study concludes that well-designed shear transfer systems are critical to achieving full composite behaviour and highlights design considerations for optimizing structural performance in floor systems, beams, and wall panels. These results contribute to the development of reliable design guidelines for cold-formed steel–concrete composite construction.

Keywords: *Cold-formed steel, Composite construction, Steel–concrete interaction, Shear transfer, Embossments, Shear connectors, Local buckling, Flexural strength, Load-carrying capacity, Finite-element analysis, Ductility, Concrete confinement, Structural performance, Floor systems, Wall panels.*



MSEC-RS-2025-DI-004

Dual Gold Nanowire-Based D-Shaped PCF-SPR Sensor for Enhanced Sensitivity and Wide-Range Refractive Index Detection

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Abstract

A novel D-shaped PCF-SPR sensor integrating dual gold nanowires is introduced to achieve enhanced sensitivity and broad-range refractive index monitoring. The device enables simultaneous measurement of refractive index (1.20–1.40) and temperature (–20 to 60 °C), ensuring versatile performance for diverse sensing environments. It exhibits remarkably high wavelength sensitivities of 51,200 and 56,700 nm/RIU, along with strong amplitude sensitivities of –98.9 and –147.6 RIU⁻¹. The sensor also provides a temperature sensitivity of 17.4 nm/°C. Numerical simulations conducted using FEM in COMSOL Multiphysics, supported by MATLAB-based data analysis, validate the design's efficiency and stability. Owing to its precise detection capability and reliable response, the proposed structure shows strong potential for biochemical, industrial, and environmental sensing application.

Keywords: *D-shaped PCF-SPR sensor, Dual gold nanowires, Refractive index sensing, Temperature sensitivity, FEM simulation (COMSOL)*



MSEC-RS-2025-HR-005

Geotechnical Investigation on Dredged Marine Sand - A Review

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Abstract

Dredged marine sand is increasingly utilized in various geotechnical applications such as land reclamation, foundation construction, and coastal engineering projects. However, its unique characteristics pose challenges for geotechnical investigation and design. This review synthesizes current research on the geotechnical properties of dredged marine sand, focusing on its physical, mechanical, and hydraulic behaviour. Key aspects covered include grain size distribution, mineralogy, density, shear strength, compressibility, and permeability. The review highlights the influence of dredging and placement processes on sand characteristics, as well as the importance of accurate site investigation techniques such as sampling, in-situ testing, and laboratory analysis. Challenges related to variability, contamination, and consolidation behaviour are discussed, along with advances in testing methods and stabilization techniques. The findings emphasize the need for comprehensive investigation protocols to ensure the safe and effective use of dredged marine sand in engineering projects. This review provides a valuable reference for researchers, engineers, and practitioners involved in marine geotechnical engineering.

Keywords: *Dredged marine sand, geotechnical investigation, contamination, mineralogy, stabilization techniques, soil characteristics*



MSEC-RS-2025-AI&DS-001

AI-Based Decision Support System

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Abstract

Decision Support Systems (DSS) are crucial instruments that help businesses evaluate complicated data and arrive at well-informed decisions. Traditional DSS approaches have limits in terms of scalability, adaptability, and real-time processing due to the rapid expansion of data across domains. In order to improve decision-making, this research suggests an AI-Based Decision Support System that makes use of machine learning, data analytics, and optimization strategies. The system continuously learns from data, seeks to offer predictive insights, suggest the best course of action, and adjust to changing surroundings. The domain-independent design, the suggested framework can be tailored for a wide range of applications, including smart cities, healthcare, energy management, and finance. The system guarantees openness and confidence in the decision-making process by including explainable AI techniques. The anticipated result is a strong and intelligent DSS that increases decision accuracy, efficiency, and dependability in challenging and unpredictable situations.

Keywords: *AI-Based DSS, Machine Learning, Predictive Analytics, Explainable AI*



MSEC-RS-2025-AI&DS-002

5G/6G-Enabled Cyber-Physical Systems: Architecture, Challenges, and Intelligent Connectivity for Next-Generation Automation

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Abstract

Cyber-Physical Systems (CPS) represent tightly integrated networks of computation, communication, and physical processes. With the advent of advanced wireless technologies such as 5G and upcoming 6G networks, CPS are evolving toward ultra-reliable, intelligent, and large-scale connected ecosystems. This paper explores how 5G and 6G technologies transform CPS by enabling ultra-low latency, massive device connectivity, high bandwidth, and AI-driven network intelligence. Key enablers such as edge computing, network slicing, terahertz communication, and AI-native architecture in 6G significantly enhance real-time monitoring, control, and decision-making in critical CPS applications. These include Industry 4.0 automation, autonomous transportation, smart healthcare, smart grids, and mission-critical systems. The paper presents the layered architecture of 5G/6G-enabled CPS, discusses major challenges related to security, interoperability, scalability, and reliability, and highlights future research directions for building resilient and autonomous systems. This study concludes that 5G and 6G technologies will be foundational pillars in developing the next generation of intelligent, adaptive, and self-optimizing Cyber-Physical Systems.

Keywords: *5G Networks, 6G Communication, Cyber-Physical Systems (CPS), Ultra-Reliable Low-Latency Communication (URLLC), Intelligent Edge Computing, Network Slicing, Terahertz Communication, AI-Native Networks, Industrial Automation, Smart Citi*



MSEC-RS-2025-AI&DS-003

Construction Safety AI

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Abstract

Construction sites are among the most hazardous work environments, with a high risk of injuries and fatalities due to non-compliance with safety protocols—particularly the improper use or absence of Personal Protective Equipment (PPE). This project presents Construction Safety AI, an intelligent monitoring system designed to enhance workplace safety through automated detection of PPE violations in real-time. Leveraging the capabilities of the YOLOv11 object detection model, the system accurately identifies instances where workers fail to wear essential safety gear, including hardhats, masks, and safety vests. The system architecture is built using Flask for both the frontend and backend, ensuring a lightweight yet robust web-based interface. It supports multiple modes of video input, including uploaded footage, live webcam streams, and CCTV video feeds, making it adaptable to diverse construction environments. Upon detection of a violation, the system triggers an automated email alert using the Gmail API, allowing for immediate corrective action. Comprehensive testing and evaluation demonstrate that the system achieves high accuracy in varied lighting and environmental conditions, showcasing its reliability and real-world applicability. By automating the monitoring process, Construction Safety AI reduces the burden on manual supervisors and contributes to the creation of safer, more compliant construction sites. The project also lays the foundation for future enhancements, such as behaviour-based safety monitoring and mobile deployment.

Keywords: *PPE Detection, YOLOv11, Construction Safety, Computer Vision, Flask, AI based Alert System, Convolutional Neural Network, Deep Learnin*



MSEC-RS-2025-AI&DS-004

Assessing and Quantifying Chatbot Response Accuracy in E-Commerce using Machine Learning

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Abstract

Chatbots have become an essential component of modern e-commerce platforms, enabling automated customer support, product recommendations, and query resolution. However, their responses are not always accurate, contextually relevant, or aligned with user expectations, which negatively impacts customer trust and satisfaction. This research proposes a Machine Learning-based evaluation model to assess and quantify the accuracy of chatbot responses in real time. The study constructs a labeled dataset consisting of user queries paired with chatbot outputs and corresponding human-validated accuracy annotations. Multiple supervised learning algorithms are examined, and the BERT (Bidirectional Encoder Representations from Transformers) model is selected due to its superior ability to understand semantic context and linguistic nuances. The proposed system processes each chatbot interaction and generates an accuracy score based on contextual relevance, completeness, and correctness of the response. This score can be integrated into e-commerce platforms to enable continuous monitoring, performance improvement, and adaptive fine-tuning of chatbot models. Experimental results demonstrate that the BERT-based evaluation system provides robust and reliable accuracy assessments, ultimately enhancing customer experience and increasing the reliability of automated conversational agents.

Keywords - *Machine Learning; Chatbot Evaluation; BERT; Natural Language Processing; E-commerce Automation; Response Accuracy Scoring; Supervised Learning; Contextual Analysis; Conversational AI; Customer Satisfaction*



MSEC-RS-2025-AI&DS-005

Lung Cancer Detection using Faster R-CNN

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Abstract

Early detection of pulmonary nodules is essential for timely lung cancer diagnosis and improved patient survival. However, traditional Computer-Aided Detection (CAD) systems often face challenges such as high false-positive rates and difficulties handling nodules of varying sizes. This study introduces a Faster R-CNN-based model for detecting lung nodules in CT scans. The proposed approach improves lung cancer screening by delivering more accurate nodule identification, reducing false positives, and supporting earlier diagnosis.

Keywords: *Pulmonary nodules, Lung cancer, CT scans, Computer-Aided Detection (CAD), Faster R-CNN, Early diagnosis*



MSEC-RS-2025-AI&DS-006

Impact of Artificial Intelligence on Informed Business Decisions Using Social Media Apps in India: A TOE Framework Analysis

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Abstract

This study investigates how Indian businesses use artificial intelligence (AI) tools integrated with social media platforms to enhance decision-making. Using the Technology-Organization-Environment (TOE) framework, the research identifies the major factors influencing AI adoption for social-media-driven business intelligence in the Indian market. A quantitative method was used, with structured questionnaires distributed to 400 business professionals across multiple industries and enterprise sizes in major metropolitan cities. Regression analysis was applied to evaluate relationships between TOE variables and business decision-making performance. Results show that businesses using AI-powered social media analytics achieved 28% more informed strategic decisions and 33% better market responsiveness compared to non-adopters. The study acknowledges limitations such as its cross-sectional design and metropolitan-focused sample, which may not fully represent rural business contexts. Future longitudinal studies should examine AI implementation effectiveness across diverse regions and cultural settings in India. Overall, the findings provide a decision-making framework for AI adoption by Indian businesses, highlighting the importance of AI training, strong data governance, strategic alignment, and contextualizing global AI solutions for India's unique market dynamics.

Keywords: *Artificial Intelligence, Technology-Organization-Environment (TOE) Framework, Business Intelligence*



MSEC-RS-2025-AI&DS-007

FLAVA: Federated Learning Audio Voice Assistant for Privacy- Preserving Speech Recognition

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Abstract

Voice assistants play a central role in modern human–computer interaction by enabling hands-free operation across smartphones, smart speakers, wearables, and embedded systems. Their ability to interpret and respond to spoken commands has reshaped expectations for seamless and intuitive digital experiences. However, most current voice assistant architectures rely on centralized processing pipelines where raw audio or extracted features are transmitted to remote cloud servers for training and inference. This centralized model creates serious challenges, including privacy risks, potential unauthorized data access, and reliance on reliable network connectivity. The accumulation of sensitive speech data in central servers also raises ethical and regulatory concerns related to confidentiality and evolving data-protection standards. To overcome these limitations, this paper introduces FLAVA (Federated Learning Audio Voice Assistant), a privacy-preserving framework that decentralizes the learning process. FLAVA uses Federated Learning, allowing distributed model training directly on user devices so that raw audio never leaves local storage. Instead of sharing voice recordings, FLAVA transmits only model updates that are securely aggregated to create a global model capable of improving continuously while maintaining user privacy.

Keywords: *Federated Learning, Privacy-Preserving Voice Assistant, Edge Computing*



MSEC-RS-2025-AI&DS-008

Next-Generation Environmental Surveillance using IoT and Artificial Intelligence

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Abstract

Next-generation environmental surveillance is being reshaped by the convergence of Internet of Things (IoT) technologies and Artificial Intelligence (AI), enabling continuous, real-time monitoring of critical environmental parameters. Traditional monitoring methods often lack spatial coverage, responsiveness, and predictive capabilities, resulting in delayed detection of hazards and limited situational awareness. IoT-based sensor networks, supported by low-power communication technologies and edge computing, facilitate large-scale acquisition of air, water, soil, and climate data with high temporal resolution. AI-driven analytics—including machine learning, deep learning, and geospatial intelligence—transform this data into actionable insights by enabling automated anomaly detection, pollution forecasting, and early warning generation for events such as floods, forest fires, chemical leaks, and disease outbreaks. Cloud–edge architectures ensure scalable processing and efficient data management, while digital twins allow simulation and decision support for environmental planning. Despite technological progress, challenges persist related to data quality, cybersecurity, system interoperability, and sustainable sensor deployment. This paper highlights the technological advancements, applications, and limitations of AI–IoT integration and outlines future directions for building intelligent, resilient, and sustainable environmental monitoring systems.

Keywords: *Artificial Intelligence, Environmental Surveillance, Deep Learning, Machine Learning, IoT Sensors, Remote Sensing, Satellite Imagery, Predictive Analytics, Environmental Monitoring, Anomaly Detection.*



MSEC-RS-2025-AI&DS-009

A Real-Time Safety and Trust Alignment Framework for Autonomous AI Agents

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Abstract

The rapid deployment of autonomous AI agents in domains such as healthcare, transportation, manufacturing, and smart environments demands strict guarantees of safety, transparency, and trust. However, current AI systems often operate as black boxes, making it difficult for humans to predict, verify, or align their behaviour in real time. This research proposes a Real-Time Safety and Trust Alignment Framework (RT-STAF) designed to ensure reliable, explainable, and human-aligned operation of autonomous agents. The framework integrates real-time safety monitoring using predictive risk analysis and constraint-based control to detect unsafe or abnormal agent behaviour, adaptive trust modelling that dynamically adjusts autonomy levels based on user confidence, system performance, and environmental uncertainty, and transparent decision explanation using interpretable AI techniques to provide human-understandable justifications for agent actions. RT-STAF aims to create a continuous feedback loop between humans and AI agents, enabling fast intervention, improved reliability, and enhanced accountability. This framework supports the development of safe, trustworthy, and ethically guided autonomous systems suitable for deployment in safety-critical real-world applications.

Keywords: *Autonomous AI Agents, Real-Time Safety Monitoring, Trust Alignment, Explainable AI, Human–AI Interaction, Adaptive Trust Models, Safety-Critical Systems, AI Reliability, Ethical AI, Intelligent Autonomous System*



MSEC-RS-2025-AI&DS-010

Curiosity-Driven Reinforced Learning of Undesired Actions in Autonomous Intelligent Agents

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Abstract

Curiosity-driven reinforcement learning enables autonomous agents to explore environments more efficiently by rewarding novelty and self-supervised discovery. However, this exploration mechanism can cause agents to unintentionally learn unsafe or undesirable actions, particularly in environments with sparse or ambiguous rewards that do not align with human intentions, creating risks in real-world domains such as robotics, autonomous vehicles, industrial automation, and service AI. This research examines how curiosity-driven policies lead to the emergence and persistence of harmful behaviours and proposes a framework to mitigate these effects by analysing the internal learning dynamics that inflate intrinsic rewards for irrelevant or unsafe actions and evaluating how uncertainty estimation, novelty detection, and intrinsic reward shaping affect exploration. To address these challenges, a safety-aware curiosity modulation strategy is introduced, integrating constrained reinforcement learning, behaviour filtering, and human-aligned reward correction to ensure that exploration remains productive, safe, and goal-directed. The approach maintains the benefits of curiosity-driven exploration while preventing reinforcement of actions that conflict with safety or task objectives.

Keywords: *Curiosity-Driven Learning; Reinforcement Learning; Autonomous Intelligent Agents*



MSEC-RS-2025-AI&DS-011

Transformers and their impact on Natural Language Processing

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Abstract

This project presents an AI-based natural language processing system built using Transformer architecture. The system leverages attention mechanisms to understand context, meaning, and relationships within text. The model is trained/fine-tuned on domain-specific data to perform tasks such as text classification, question answering, or summarization. The Transformer-based approach provides superior accuracy compared to traditional RNN/LSTM models due to its ability to process input in parallel and capture long-range dependencies. The project includes preprocessing, tokenization, model evaluation, and real-time user interaction through a simple interface. The system demonstrates efficient language understanding and offers scalable applications in education, customer service, and automation.

Keywords: *RNN/LSTM, token, Mode*



MSEC-RS-2025-AI&DS-012

A Novel Approach to Explainable Artificial Intelligence (XAI) in Healthcare

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Abstract

The increasing adoption of artificial intelligence (AI) in healthcare has highlighted the need for transparent, trustworthy, and clinically interpretable decision-making systems. This paper presents a novel approach to Explainable Artificial Intelligence (XAI) that integrates model-agnostic interpretability methods with domain-specific clinical reasoning frameworks. Our approach combines layered explanations—global model insights, case-based reasoning, and patient-level visual or textual justifications—to provide clarity across multiple user groups, including clinicians, patients, and healthcare administrators. By incorporating medical ontologies and causal inference techniques, the framework enhances explanation relevance, reduces ambiguity, and aligns model outputs with established clinical knowledge. Experimental results on real-world healthcare datasets demonstrate improved diagnostic accuracy while maintaining high levels of interpretability and user trust. This work contributes a scalable XAI methodology tailored for sensitive healthcare environments, supporting safer deployment of AI-driven clinical decision support systems.

Keywords: *Explainable Artificial Intelligence (XAI), Healthcare AI, Clinical Decision Support Systems, Model Interpretability, Transparency, Medical Ontologies, Causal Inference, Trustworthy AI, Machine Learning, Explain ability Framework, Patient-Level Explanations, Model-Agnostic Methods.*



MSEC-RS-2025-AI&DS-013

Leveraging Deep Learning for Predictive Modeling in Clinical Research and Pharmaceutical Discovery

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Abstract:

The traditional drug discovery process is notoriously slow, expensive, and inefficient, with nearly 90% of candidates failing after entering phase-I clinical trials. This paper argues that deep learning (DL) represents a fundamental paradigm shift, offering a transformative solution to these persistent challenges. We demonstrate how DL frameworks are revolutionizing core tasks, from accelerating disease diagnosis and target identification to optimizing virtual screening and predicting drug toxicity with greater accuracy. By leveraging models that learn directly from complex, high-dimensional data—including genomic sequences, molecular structures, and electronic health records—DL uncovers critical patterns that elude conventional methods. The primary impact is a significant de-risking of the therapeutic pipeline, enhancing the probability of success while drastically reducing development timelines. We conclude that the integration of these advanced computational tools is essential for fostering innovation and realizing the future of data-driven, personalized healthcare.

Keywords: *Deep Learning (DL), Drug Discovery, Virtual Screening, Target Identification, Disease Diagnosis, Drug Toxicity Prediction, High-Dimensional Biomedical Data, Genomic Sequences, Molecular Structures, Electronic Health Records (EHR), Therapeutic Pipeline, Personalized Healthcare, Computational Drug Design, AI in Healthcare.*



MSEC-RS-2025-CIVIL-001

Performance Analysis of Buildings

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Abstract

This paper presents a comparative analysis between the design of a sustainable building and a non-sustainable counterpart, highlighting the architectural, environmental, and economic differences. The sustainable model incorporates eco-friendly materials, energy-efficient systems, and passive design strategies aimed at minimizing environmental impact and operational costs. In contrast, the non-sustainable building reflects conventional construction practices with limited regard for long-term ecological consequences. Through this dual design approach, the project emphasizes the critical role of sustainability in modern architecture and advocates for responsible building practices that align with global environmental goals. We have reduced the consumption of the U-Value for walls, roof, floor and windows in Sustainable building over Non-Sustainable building as nearly 40%. Also optimized energy intensity of nearly 50kWh/m³year and other optimization for both the buildings are described in the detailed way.

Keywords: *Sustainable, energy, building, material, environmental, performance*



MSEC-RS-2025-CIVIL-002

Study of Soft Storey with Irregular Configuration of Multistorey Building in Various Seismic Zone

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Abstract

The seismic performance of multi-storey buildings is significantly influenced by the uniformity and regularity of their structural configuration in both plan and elevation. Among various structural irregularities, the presence of a soft storey, a storey having considerably lower stiffness or strength compared to adjacent storeys has been recognized as a major factor contributing to severe damage or collapse during earthquakes. This study examines the seismic behaviour of multi-storey reinforced concrete (RC) buildings incorporating soft storey irregularities under different seismic zone conditions, following the guidelines of IS 1893 (Part 1): 2016. The analysis aims to evaluate the effects of soft storey formation on lateral displacement, inter-storey drift, base shear, and overall structural stability, thereby contributing to better understanding and safer design practices for irregular buildings in earthquake-prone regions.

Keywords: *Seismic irregularity, Soft storey, Seismic performance, Inter-storey drift, Base Shear.*



MSEC-RS-2025-CIVIL-003

Simulation-Driven Urban Mobility: A Data Science Approach using Simio and Python

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Abstract

The increasing complexity of urban transportation systems demands analytical and simulation-based approaches that move beyond intuition-driven planning. This research presents a framework titled “Simulation-Driven Urban Mobility: A Data Science Approach using Simio and Python,” which integrates discrete-event simulation with data-driven modeling for improved mobility decision-making. In this study, Simio is employed to simulate cyclic urban transport operations—such as ready-mix concrete truck dispatching, public transport routing, and travel demand cycles—capturing system behavior under varying resource and route configurations. The simulation outputs, including travel time, queue length, and resource utilization, are exported and analyzed using Python-based data science techniques. Libraries such as Pandas, Scikit-learn, and Matplotlib are used to perform regression, clustering, and optimization analyses to predict performance indicators and identify efficiency patterns across transport scenarios. This integrated methodology bridges the gap between civil engineering simulation and machine learning analytics, enabling engineers to test hypotheses, validate assumptions, and optimize transport operations quantitatively. The results demonstrate how combining simulation with data science enhances predictive accuracy, reduces operational delays, and supports evidence-based decision-making for sustainable urban mobility.

Keywords: *Urban Mobility Simulation, Discrete-Event Modeling (Simio), Data-Driven Transport Analytics, Machine Learning Integration, Sustainable Urban Transportation Planning*



MSEC-RS-2025-CIVIL-004

OMNI-Tank: A Single-Shell EC-MBR for Sustainable Wastewater Treatment & Reuse

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Abstract

The rapid growth of urban populations and industrial activities has intensified the need for compact, energy-efficient and sustainable wastewater treatment systems capable of delivering high-quality effluent suitable for reuse. Conventional decentralized treatment technologies often require multiple treatment units, larger footprints, and complex operational protocols, which limit their feasibility in resource-constrained settings. To address these challenges, the present study introduces the Omni-Tank, an integrated Single-Shell Electrocoagulation–Membrane Bioreactor (EC-MBR) system designed to unify physical, chemical, and biological treatment mechanisms within a single compact unit. The system merges three treatment zones - a reaction chamber, an electrocoagulation assembly, and a submerged membrane module - into one enclosed shell, significantly reducing space, energy consumption, and operational complexity. This mechanism enhances suspended solids removal and reduces membrane fouling, thereby improving the overall efficiency of the downstream MBR stage. The membrane bioreactor integrates a high-activity activated sludge process with a hollow-fiber microfiltration module that ensures the removal of biomass, pathogens, and fine particulates. By leveraging the synergistic benefits of EC pretreatment and membrane filtration, the Omni-Tank achieves superior pollutant removal with reduced aeration requirements and extended membrane lifespan.

Keywords - *Electrocoagulation (EC), Membrane Bioreactor (MBR), Integrated Wastewater Treatment, Single-Shell Reactor, Sustainable Water Reuse, Omni-Tank*



MSEC-RS-2025-CIVIL-005

Self-Curing Concrete

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Abstract

Concrete is the foundation of modern construction, but its curing process presents noteworthy contests. Conventional curing methods, such as water spraying or covering with wet sac bags require periodical monitoring, and may not be effective under extreme conditions, such as high temperatures or low humidity. This is where self-curing concrete offers a right solution. By incorporating specific materials or chemicals, self-curing concrete ensures that moisture is retained during the curing process, allowing for enhanced hydration and better recital of the concrete. This study explains the role of self-curing chemicals namely poly ethylene glycol (PEG). After making cylinders and cubes, different tests were done to confirm the compressive strength, tensile strength for 3, 7, 28 days were carried out. Poly ethylene glycol has done extremely better internal curing.

Keywords: *Internal Curing, Polyethylene Glycol (PEG), Lightweight Aggregates, Superabsorbent Polymers (SAPs), Reduced Shrinkage, Improved Hydration*



MSEC-RS-2025-CIVIL-006

Magnetically Responsive Concrete Incorporating Nano-Ferrits for Crack Resistance and Durability Enhancement

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Abstract

Concrete, the most widely used construction material, is inherently brittle and prone to cracking due to thermal stresses, shrinkage, and external loads. These cracks compromise structural integrity, durability, and service life. The present study explores the development of Magnetically Responsive Concrete (MRC) incorporating nano-ferrite particles to enhance crack resistance and durability. Nano-ferrites, known for their magnetic and self-sensing capabilities, are dispersed within the cement matrix to impart smart and adaptive behavior to the composite. When subjected to magnetic fields, these particles align and influence the microstructure, leading to improved bond strength, reduced porosity, and enhanced crack-healing potential through particle migration and reorientation at the microcrack interfaces. Experimental investigations include mechanical strength tests, microstructural analysis, and durability assessments such as water permeability and chloride penetration resistance. The results indicate a significant improvement in compressive and tensile strength, reduced crack width, and enhanced self-healing capacity compared to conventional concrete. The study demonstrates that magnetically responsive concrete with optimized nano-ferrite content can provide a sustainable, self-adaptive material for long-lasting infrastructures.

Keywords: *Magnetically Responsive Concrete, Nano-Ferrite Particles, Self-Healing, Crack Resistance, Durability, Smart Materials.*



MSEC-RS-2025-CIVIL-007

Structural Performance on Self-Consolidating Ultra-High-Performance Fiber-Reinforced Concrete

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Abstract

Self-Consolidating Ultra-High-Performance Fiber-Reinforced Concrete (SCHPFRC) represents a significant advancement in concrete technology, offering a combination of enhanced workability, exceptional mechanical properties, and superior durability. By integrating self-consolidating concrete's ability to flow and fill formwork without mechanical vibration with the extraordinary strength and toughness of ultra-high-performance concrete (UHPC), along with fiber reinforcement, SCHPFRC provides an optimized material solution for structural applications in modern construction. This study focuses on evaluating the structural performance of SCHPFRC through comprehensive experimental and analytical investigations. The research examines critical performance metrics, including compressive strength, tensile strength, flexural capacity, ductility, and energy absorption under static and dynamic loading conditions. The addition of fibers significantly improves post-cracking behavior, enhances deformation capacity, and delays crack propagation, which is crucial.

Keywords: *High-performance concretes, Fiber-Reinforced Concrete, Ultra-High-Performance, Self-Consolidating, energy absorption.*



MSEC-RS-2025-CIVIL-008

Performance Evaluation of Smart Concrete by the Addition of Nanosilica and Nano TiO₂ – A State of the Art Review

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Abstract

In recent years, enhancing the performance and durability of concrete through nanotechnology has attracted significant interest in sustainable construction. This state-of-the-art review comprehensively examines the roles of nano-silica (nSiO₂) and nano-titanium dioxide (nTiO₂) as functional additives in smart concrete, with emphasis on their effects on mechanical, durability, microstructural, and multifunctional properties. Key mechanisms by which nSiO₂ improves concrete behavior—such as its high pozzolanic reactivity, filler effect for pore refinement, nucleation of calcium-silicate-hydrate (C-S-H) gel, reduction in calcium hydroxide content, and densification of the interfacial transition zone—are discussed in detail. Similarly, the review explores the contribution of nTiO₂ not only to mechanical strength and durability (e.g., resistance to chemical attack, permeability, chloride ingress) but also to its photocatalytic, self-cleaning, and environmental pollutant degrading functionalities. Critical factors such as optimal dosage, dispersion techniques, particle size and morphology, interaction with supplementary cementitious materials, effects on fresh state properties (workability, setting time), long-term performance, and possible trade-offs are analyzed. concrete performance and sustainability, but their effective use requires precise control of material properties and mix design.

Keywords: *Sustainable construction, nano silica, nano tio2, durability, microstructural property, long term performance*



MSEC-RS-2025-CIVIL-009

A Study on Reactive Powder Concrete (RPC) and Prediction using Artificial Neural Network (ANN)

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Abstract

Reactive Powder Concrete (RPC) is an advanced ultra-high-performance cementitious composite designed to achieve superior strength, durability, and micro-structural homogeneity. By eliminating coarse aggregates and optimizing particle packing through fine powders like silica fume and quartz, RPC attains compressive strengths exceeding 150 MPa with remarkable ductility when reinforced with steel fibers. This study experimentally investigates the mechanical properties of RPC and predicts its compressive strength using an Artificial Neural Network (ANN) model. The experimental program involves testing cube and prism specimens for compressive, flexural, and split tensile strengths at 7, 14, and 28 days. The ANN model is trained using mix proportion parameters—cement content, silica fume ratio, water-binder ratio, and steel fiber percentage—to predict compressive strength.

Keywords: *RPC, Compressive concrete, steel fibre and ANN*



MSEC-RS-2025-CIVIL-010

Experimental Investigation and Parametric Studies on Compressive Strength of Geopolymer Concrete

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Abstract

The demand of concrete is increasing day by day for satisfying the need of development of infrastructure facilities. It is well established fact that the production of OPC not only consumes significant amount of natural resources and energy but also releases substantial quantity of carbon dioxide to the atmosphere. Therefore, it is essential to find alternatives to make the concrete environment friendly. Geo polymer is an inorganic alumino - silicate compound, synthesized from fly ash. The fly ash, one of the source materials for geo polymer binders, is available abundantly in India ,but to date its utilization is limited, Hence it is essential to make the efforts to utilize this by product in concrete manufacturing in order to make the concrete more environmental friendly. This paper describes the experimental work have conducted by casting 20 geo polymer concrete mixes to evaluate the effect of various parameters affecting it's the compressive strength in order to enhance its overall performance.

Keywords: *Geopolymer Concrete, Compressive Strength, Alumino-Silicate*



MSEC-RS-2025-CIVIL-011

Utilization of Non-Biodegradable Waste Materials in Bituminous Mixes for Sustainable Road Construction

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Abstract

This study discussed the suitability of plastic waste materials for pavement construction... The waste is mixed in different proportions to the soil sample and their influences on geotechnical properties were studied. The results of the tests indicated that plastic alone is not suitable for pavement sub grade. When quarry dust was added along with soil plastic mix, it maintains the CBR value within the required range. Plastic use in road construction is not new. It is already in use as PVC or HDPE pipe mat crossings built by cabling together PVC (polyvinyl chloride) or HDPE (high-density poly-ethylene) pipes to form plastic mats. Plastic roads. Plastic roads mainly use plastic carry-bags, disposable cups and PET bottles that are collected from garbage dumps as an important ingredient of the construction material. When mixed with hot bitumen, plastics melt to form an oily coat over the aggregate and the mixture is laid on the road surface like a normal tar road. Plastic roads would be a boon for India's hot and extremely humid climate, where temperatures frequently cross 50°C and torrential rains create havoc, leaving most of the roads with big potholes. The use of the innovative technology not only strengthened the road construction but also increased the road life as well as will help to improve the environment and also creating a source of income

Keywords: *Coarse aggregate, bitumen, quarrydust, shredded plastic*



MSEC-RS-2025-CIVIL-012

A Feasibility Study of Different Concrete Ratios for 3D Mix

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Abstract

3D printable concrete is a special type of concrete that can be used for fast-track construction. Many research studies have been conducted on various aspects of 3D printing technology. It can also be noted as additive manufacturing. Under micro prototyping using 3D filament, special types of plastics called thermoplastics are used. It uses extremely large-scale 3D printers, which measure several meters in height and length, to extrude some concrete from the nozzle. The benefits include error reduction, which improves safety outcomes for workers. The course of extrusion-based concrete printing requires the printable mix to flow through the network and be stored in layer by layer directly on site, analogues to the ink jet printing process. The printing process requires the (type of the pump, size of the nozzle and extrusion velocity) to study. From the study, it was found that both extrudability and durability can we found only when the yield stress was within the range of 1.5-2.5kPa. As per the previous research, mixes are prepared with rice husk ashes with viscosity viscosity-modifying agent performance under the structural build-up will be studied.

Keywords: *3D printable concrete, extrusion-based printing, mortar ratio, printability, yield stress*



MSEC-RS-2025-CIVIL-013

Performance Evaluation of Reactive Powder Concrete with Partial Replacement of Natural Fine Aggregate by Recycled Fine Aggregate

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Abstract

The study aims to evaluate the mechanical properties and durability of RPC with different proportions of RA replacement. The mix design of RPC is optimized by using silica fume and superplasticizer to enhance its strength and workability. The RA is incorporated in the mix to reduce the environmental impact of concrete production and waste management. The experimental results show that the compressive and flexural strengths of RPC decrease with an increase in RA replacement, but the impact resistance and chloride ion permeability improve. The study suggests that a suitable combination of RPC and RA can result in a sustainable and durable construction material with adequate mechanical properties for various applications. The use of Recycled Aggregate (RA) in Reactive Powder Concrete (RPC) has gained significant attention due to its sustainability and economic advantages. This paper aims to investigate the feasibility of using RA as a partial replacement of natural coarse aggregate in RPC.

Keywords: *RPC, Mix proportion, silica fume, superplasticizer.*



MSEC-RS-2025-CSE-001

Detection of Fraud in Financial Transactions

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Abstract

In today's digital era, online banking has become indispensable, enabling quick and convenient financial transactions. With the widespread adoption of credit cards for payments, there has also been a parallel increase in fraudulent or unauthorized credit card activities. Detection of such illegal transactions is essential to minimize financial losses for banking institutions. A major limitation of existing fraud detection systems is their tendency to misclassify genuine transactions as fraudulent, resulting in unwarranted transaction declines, monetary loss, and customer dissatisfaction. Therefore, there is a need for a more reliable and effective approach to detect fraudulent credit card usage. This paper presents a hybrid model that integrates the Random Forest algorithm with Logistic Regression to improve the prediction of illegitimate credit card transactions. The proposed approach is assessed using accuracy and additional performance metrics such as F1-score, recall, and precision.

Keywords: *Logistic Regression, Machine Learning, SMOTE, Random Forest, Fraudulent Card Transactions.*



MSEC-RS-2025-CSE-002

Oncocrisprai: AI-Driven Optimization of CRISPR-Based Gene-Editing Strategies for Precision Oncology

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Abstract

Precision oncology requires gene-editing strategies tailored to each tumors' molecular landscape. CRISPR-based genome engineering offers transformative possibilities but demands precise guide. This research lies at the intersection of artificial intelligence, computational biology, and precision oncology—focusing on the use of AI to optimize CRISPR-based gene-editing for personalized cancer treatment. It combines machine learning, systems biology, and omics integration to enable context-aware genomic interventions. Existing CRISPR design frameworks like Deep CRISPR and CRISPR-Net emphasize sequence-based prediction and efficiency scoring but lack integration of tumour-specific omics data and dynamic feedback mechanisms. Their static design models limit adaptability to diverse cancer subtypes. Current methods struggle with high false-positive off-target predictions, inadequate modelling of gene-network interactions, and poor clinical interpretability. The absence of adaptive learning reduces accuracy and limits translational potential in oncology research. The system generates ranked, experimentally actionable CRISPR strategies and clinician-friendly validation reports, significantly reducing manual screening time. It enables adaptive learning from experimental feedback, improving model accuracy and translational readiness. Compared to existing approaches, OncoCRISPRAI achieves an approximate 25–30% improvement in prediction accuracy, 70% reduction in off-target error rates, and over 50% enhancement in context-specific guide selection precision based on preliminary validation datasets.

Keywords: *Precision Oncology, CRISPR Gene Editing, AI-Driven Genome Engineering, Multi-omics Integration, Graph Neural Networks, Personalized Cancer Therapy*



MSEC-RS-2025-CSE-003

Fish Classification using Machine Learning Algorithms

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Abstract

Fish classification plays a vital role in marine research, aquaculture management, and automated monitoring systems. Traditional manual identification is slow, error-prone, and depends heavily on expert knowledge. To address these limitations, this work applies Machine Learning (ML) and Deep Learning (DL) techniques for automatic fish species classification using image data. The study begins with the collection of labeled fish images from public datasets and field recordings. The images undergo preprocessing such as resizing, normalization, and augmentation to handle variations in lighting, pose, and underwater noise. Two categories of ML approaches are explored. The classical ML methods using hand-crafted features (HOG, LBP, color descriptors) with classifiers like SVM and Random Forest, and deep learning based methods, primarily Convolutional Neural Networks (CNNs). Transfer learning with MobileNetV2 is adopted to improve accuracy and reduce training time. The pre-trained CNN extracts high-level features automatically, and a custom dense layer is added for species classification. Performance is measured using accuracy, precision, recall, F1-score, and a confusion matrix. Results show that deep learning significantly outperforms classical methods, achieving high classification accuracy due to its strong feature-learning capability. The model is robust against variations in fish appearance and environmental conditions.

Keywords: *Fish Classification, Machine Learning, Deep Learning, Convolutional Neural Networks (CNN), Transfer Learning, MobileNetV2*



MSEC-RS-2025-CSE-004

Safe Stride: Simulation-Assisted Obstacle Detection and Navigation Aid for Multi-Directional Mobility

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Abstract

Assistive technologies are essential for supporting safe and independent mobility among visually impaired individuals. This paper presents SafeStride, a wearable, multi-directional obstacle detection and navigation system designed to enhance situational awareness in complex environments. The system employs a tri-axis distance-sensing architecture that monitors three critical regions: the forward path at waist level, the upper zone at a 45° elevation toward the head, and the lower zone at a 45° declination toward the ground. This configuration enables continuous detection of obstacles positioned ahead, above, and below the user. Real-time sensor data are processed through an embedded control unit, which provides immediate haptic or audio feedback to facilitate timely hazard recognition and avoidance. The lightweight, clip-on design ensures unobtrusive operation and comfortable daily use. Experimental evaluation in indoor and outdoor scenarios demonstrates the system's reliability and responsiveness. By integrating multi-directional sensing with intuitive feedback, SafeStride offers an effective assistive solution that improves safety, confidence, and independent mobility for visually impaired users.

Keywords: *Assistive technology, Obstacle detection, Wearable Device, Multidirectional Sensing, Visually Impaired Navigation, Real-time Feedback*



MSEC-RS-2025-CSE-005

Approaches in Cyberbullying Detection and Prevention for Isolated Elderly Populations: A Systematic Review

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Abstract

Cyberbullying among elderly populations is an emerging yet underexplored challenge, particularly for individuals experiencing social isolation and increasing reliance on digital communication technologies. This systematic review synthesizes current research on detection and prevention approaches tailored to older adults who may be especially vulnerable due to limited digital literacy, cognitive decline, or reduced social support networks. A structured search was conducted to identify studies addressing technological, behavioral, and community-based strategies for recognizing and mitigating cyberbullying targeting seniors. Findings reveal a growing body of work focusing on machine-learning-based detection models, user-centric interface designs, educational interventions, and support frameworks integrated into social and healthcare services. However, most existing systems are not specifically adapted to the unique behavioral patterns, communication styles, or risk factors prevalent among isolated elderly individuals. The review highlights significant gaps in real-time monitoring tools that respect privacy, and multidisciplinary prevention programs. Recommendations include the development of elder-inclusive datasets, hybrid human-AI detection systems, and community-embedded outreach initiatives to enhance digital resilience. Overall, this review underscores the need for holistic, age-appropriate solutions to ensure safer online environments for isolated elderly populations.

Keywords: *Cyberbullying, Elderly Population, Social Isolation, Digital Literacy, Abuse Detection*



MSEC-RS-2025-CSE-006

Deepfake Harassment: The Next Generation of Digital Abuse

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Abstract

Deepfake harassment represents an emerging and increasingly sophisticated form of digital abuse enabled by advances in generative artificial intelligence. By manipulating or fabricating hyper-realistic audio, images, and videos, perpetrators can create non-consensual content that targets individuals—especially women and public-facing professionals—with reputational, psychological, and social harm. This next generation of abuse is characterized by low barriers to creation, high virality, anonymity of attackers, and the difficulty of verifying authenticity in online spaces. Existing legal, technical, and platform-level safeguards lag behind the speed and scale of deepfake production, creating urgent challenges for detection, governance, and victim support. This paper examines the mechanisms, threats, and societal implications of deepfake harassment while outlining multidisciplinary strategies for prevention, resilient media ecosystems, and responsible AI governance.

Keywords: *Deepfake Harassment, Digital Abuse, Generative AI, Online Safety, Misinformation, AI Governance*



MSEC-RS-2025-CSE-007

Deep Packet Inspection Meets Gradient Boosting: A Novel Approach to DDOS Attack Detection

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Abstract

Distributed Denial of Service (DDoS) attacks create a persistent threat to network infrastructure, often overwhelming systems and disrupting services. Traditional detection methods struggle to balance accuracy, speed, and adaptability in the face of evolving attack vectors. This paper presents a novel hybrid approach that integrates Deep Packet Inspection (DPI) with Gradient Boosting Machine (GBM) learning to enhance DDoS detection capabilities. DPI enables granular analysis of packet-level data, extracting rich features such as payload entropy, protocol anomalies, and traffic patterns. These features serve as input to a GBM classifier, which excels at handling complex, nonlinear relationships and imbalanced datasets. The proposed system is evaluated on benchmark datasets and real-world traffic scenarios, demonstrating superior performance in terms of detection accuracy, precision, and false positive rate compared to conventional machine learning models. Additionally, the framework supports real-time deployment, making it suitable for high-throughput environments. By combining the interpretability of DPI with the predictive power of ensemble learning, this approach offers a scalable and robust solution for proactive DDoS mitigation. The results underscore the potential of synergizing deep traffic analysis with advanced machine learning to fortify network security against increasingly sophisticated cyber threats.

Keywords: *Gradient Boosting, DDOS Detection, Deep Packet Inspection Threat Detection*



MSEC-RS-2025-CSE-008

Fake News Detection

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Abstract

The rapid proliferation of misinformation across digital platforms has intensified the need for effective fake news detection systems. Fake news, often crafted to manipulate public opinion or generate revenue, poses significant threats to social trust, political stability, and public safety. This study explores computational approaches for identifying deceptive content using a combination of natural language processing (NLP), machine learning, and network analysis techniques. Content-based methods analyze linguistic patterns, semantic inconsistencies, and stylistic cues within text, while source-based approaches evaluate publisher credibility and historical reliability. Additionally, social-context models leverage dissemination patterns and user interaction behaviors to distinguish organic information from coordinated misinformation campaigns. Recent advancements in deep learning—particularly transformer-based architectures—have substantially improved classification accuracy by capturing contextual and semantic nuances. This paper highlights current methodologies, challenges such as data scarcity and adversarial manipulation, and future directions emphasizing explainability, multimodal analysis, and real-time detection. Together, these insights contribute to the development of robust, scalable systems capable of mitigating the impact of fake news in modern information ecosystems.

Keywords: *Fake News Detection, Natural Language Processing, Machine Learning, Deep Learning Models, Network Analysis, Misinformation Identification*



MSEC-RS-2025-CSE-009

Innovating Traffic Management: Deeper Stacking-Based Group Method for Time-Sensitive Traffic Speed Predictions

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Abstract

In this scenario, optimising traffic flow via intelligent technologies is critical. As urbanisation grows, so does the requirement for efficient transportation systems. This study offers a unique technique, known as the Deeper Stacking-Based The Group Method (DSBGM), that is meant to handle the special issues associated with limited period traffic speed control. The proposed method combines sophisticated stacking techniques with deep learning architectures in order to improve the precision as well as dependability of traffic speed estimates over time intervals. The DSBGM uses a large dataset with temporal and geographical characteristics to train a deep stack structure capable of identifying subtle patterns and relationships in the traffic system. The model is developed via an iterative stacking method to obtain maximum prediction performance, especially in settings with time-sensitive fluctuations in traffic speed. Furthermore, the article examines the DSBGM's practical consequences, emphasising its potential to enhance traffic control tactics, decreasing congestion, and increasing overall transportation efficiency.

Keywords: *Traffic speed prediction, Deep Learning, Stacking-Based Model (DSBGM), Intelligent Transportation systems, Urban traffic optimization*



MSEC-RS-2025-CSE-010

Deep Fake Detection using Deep Learning

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Abstract

This comprehensive study delves into the dynamic landscape of deep learning applications, focusing on the burgeoning realm of deep fakes. Deep learning has seamlessly integrated into fields like natural language processing, machine learning, and computer vision, giving rise to innovative applications. However, the surge in deep fakes, sophisticatedly manipulated videos/images, has become a pressing concern. Existing CRISPR design frameworks like Deep CRISPR and CRISPR-Net emphasize sequence-based prediction and efficiency scoring but lack integration of tumour-specific omics data and dynamic feedback mechanisms. Their static design models limit adaptability to diverse cancer subtypes. The nefarious applications of this technology, such as fake news, celebrity impersonations, financial scams, and pose significant threats in the digital realm. Particularly, public figures like celebrities and politicians are highly susceptible to the Deep fake detection challenge. This research systematically assesses both the production and detection aspects of deep fakes, employing diverse deep learning algorithms, including InceptionResnetV2, VGG19, CNN, and Xception. The evaluation, conducted on a Kaggle deep fake dataset, highlights Xception as the most accurate among the algorithms studied. As malicious uses of deep fakes escalate, the imperative for robust detection mechanisms intensifies to safeguard against potential societal consequences.

Keywords: *Deepfake detection, Deep learning, Convolutional neural networks*



MSEC-RS-2025-CSE-011

Word Level Classification of Dravidian Languages

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Abstract

Language detection is the process of automatically identifying the language used in a text, even when that text is not always coherent or grammatically correct. Language detection has become an essential tool in today's digital landscape, where businesses rely on user-generated content to tailor advertisements, products, and services more effectively, focusing on consumers. The challenge becomes much tougher when dealing with code-mixed or multilingual text, which is common in linguistically diverse regions, such as South India. These texts often include multiple languages, sometimes written in non-native scripts, and frequently involve code-switching at various levels, making it difficult for models trained only on monolingual data to perform well. To identify the language associated with a word, a high-performance model has been proposed for the CoLI-Dravidian@FIRE 2025 shared task for word-level identification, focused on identifying Dravidian languages such as Tamil, Telugu, Malayalam, Kannada, and Tulu. The proposed system uses a language-agnostic model to identify languages associated with words using the data sets provided by the organizers of CoLI-Dravidian@FIRE 2025. The results of the proposed system are encouraging, with a Macro F1 score of 0.8995 for Kannada, 0.7434 for Tamil, 0.8271 for Malayalam, 0.9515 for Telugu, and 0.8224 for Tulu. We were ranked 1 in the leaderboard for the languages Tamil, Malayalam, and Telugu, and ranked 2 for the language Tulu. These results show the strength of the proposed model in word-level language identification of Dravidian languages.

Keywords: *Dravidian languages, Language agnostic models, Machine learning, Transformer models, Sentence embedding, Linguistic properties*



MSEC-RS-2025-CSE-012

Sip Flooding Defense System using Hybrid Detection Algorithm

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Abstract

The session initiation protocol (SIP) is the signalling protocol, for controlling voice and video communication over the internet protocol. SIP is however designed with open structure vulnerable to security attack. The SIP flooding attack is the most severe attack because it is easy to launch and capable of quickly draining the resources of both network and node. The existing flooding detection schemes are either anomaly based or misuse based. The anomaly-based scheme can detect unknown attack it does not need the prior knowledge of the attack, but it generates some false alarm, suffers from accuracy problem and gives false positive. Similarly, the misuse based schemes have high detection accuracy, no false positive but it cannot detect unknown attack. To overcome problems in both detection schemes a hybrid detection scheme is proposed. the proposed hybrid scheme consists features of both anomaly based scheme and misuse-based scheme, and it gives fast response, increase accuracy of detection and no false alarm.

Keywords: *SIP, Anomaly based detection, Misuse based detection*



MSEC-RS-2025-ECE-001

AI-Based Passenger Baggage Management System in Airport

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Abstract

The rapid growth of air travel has increased the complexity of airport baggage handling operations, demanding smarter and more efficient systems to ensure accuracy, speed, and passenger satisfaction. Traditional baggage management systems often face challenges such as misrouted, delayed, or lost baggage due to human error and limited tracking capabilities. To address these issues, this project proposes an AI-Based Passenger Baggage Management System that integrates artificial intelligence, computer vision, Internet of Things (IoT), and data analytics to automate and optimize the entire baggage handling process. The proposed system employs AI algorithms for predictive analysis and real-time decision-making, enabling proactive identification of potential delays or mishandling. Smart sensors and RFID tags facilitate continuous baggage tracking from check-in to loading and final delivery, ensuring transparency and accountability. Machine learning models analyze operational data to optimize baggage routing, improve conveyor utilization, and enhance fault detection in baggage handling equipment. Furthermore, computer vision systems assist in verifying baggage identity, detecting anomalies, and enhancing security screening accuracy. By integrating these AI-driven technologies, the system significantly reduces baggage mishandling rates, operational costs, and passenger wait times, while improving efficiency, reliability, and overall airport service quality. This intelligent approach aligns with the ICAO Airport Asset Management Framework, promoting sustainable, data-driven airport operations for the next generation of smart airports.

Keywords: *Asset management, AI, IoT*



MSEC-RS-2025-ECE-002

Smart Energy and Water Consumption Monitoring using IoT and ML

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Abstract

This Article pursuit of sustainable resource management, intelligent monitoring systems are essential for optimizing energy and water usage. This project presents a smart, IoT-based solution that enables real-time monitoring and predictive analytics for both electrical energy and water consumption. Using the ESP32 microcontroller as the core unit, the system integrates a water flow sensor and a current sensor to collect live data, which is displayed on an OLED screen for immediate feedback. The collected data is transmitted via Wi-Fi to a cloud platform for storage and further analysis. A Long Short-Term Memory (LSTM) neural network is trained using historical usage data to forecast future consumption patterns and detect anomalies such as leaks or power spikes. This predictive capability empowers users to take pre-emptive action to reduce wastage and optimize consumption. A user-friendly dashboard visualizes real-time and forecasted data, enhancing awareness and decision-making. The proposed system demonstrates a low-cost, scalable, and intelligent framework applicable to households, industries, and urban infrastructures aiming for energy and water efficiency. By combining IoT sensing with machine learning, this solution contributes to environmental sustainability and cost reduction through smarter resource utilization.

Keywords: *Machine Learning, LSTM, ESP32, Predictive Analytics, Anomaly Detection, Resource Optimization.*



MSEC-RS-2025-ECE-003

Spinal Vision CNN-Driven Techniques for Accurate Stenosis Detection

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Abstract

Spinal lumbar stenosis is a prevalent degenerative condition affecting the spinal canal, leading to compression of neural structures and subsequent symptoms such as pain and reduced mobility. Accurate and timely classification of lumbar stenosis stages is crucial for effective treatment planning and patient management. This study proposes a novel approach employing Convolutional Neural Network (CNN) techniques for the automated detection and classification of spinal lumbar stenosis stages. The proposed CNN-based framework utilizes advanced image processing and deep learning methodologies to analyse medical imaging data, such as magnetic resonance imaging (MRI) and computed tomography (CT) scans. The convolutional layers of the network are designed to automatically extract hierarchical features from the input images, capturing intricate patterns associated with different stages of lumbar stenosis. The model is trained on a diverse dataset encompassing various degrees of stenosis severity, ensuring robust generalization to unseen cases. The research focuses on achieving high accuracy, sensitivity, and specificity in classifying lumbar stenosis stages, thereby assisting healthcare professionals in making informed decisions about patient care. The results obtained from the proposed CNN-based classification system are compared with existing methods to validate its efficacy and superiority in terms of diagnostic performance. The integration of deep learning techniques in this context holds promising potential for enhancing diagnostic capabilities and streamlining treatment strategies, ultimately improving patient outcomes in the management of spinal lumbar stenosis.

Keywords: *Convolutional Neural Network (CNN), Lumbar Spinal Stenosis, Medical Image Classification, MRI/CT Analysis, Deep Learning, Diagnostic Automation*



MSEC-RS-2025-ECE-004

Acute Encephalitis Syndrome Identification using Deep Learning Method

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Abstract

Acute Encephalitis Syndrome (AES) is a rapidly progressing neurological condition that requires early diagnosis to reduce mortality and long-term disabilities. Traditional diagnostic methods—such as clinical examination, cerebrospinal fluid analysis, and neuroimaging—are time-consuming and often limited by subjective interpretation. To address these challenges, this study proposes a deep-learning-based automated system for early identification of AES using medical imaging and clinical biomarkers. Convolutional Neural Networks (CNN), hybrid CNN-LSTM models, and transfer-learning architectures (ResNet, EfficientNet) are employed to extract high-level spatial and temporal features from MRI/CT scans and time-series clinical parameters. The system is trained on preprocessed annotated datasets, utilizing data augmentation and class-imbalance handling to improve robustness. Experimental results demonstrate significant improvement in accuracy, sensitivity, and specificity compared to traditional machine-learning baselines. The proposed model can assist clinicians in faster decision-making by providing reliable, automated screening for AES, ultimately improving patient outcomes. This work highlights the potential of deep learning as a clinical support tool for neurological emergency diagnosis.

Keywords: *Acute Encephalitis Syndrome, Deep Learning, Medical Imaging, CNN-LSTM, Transfer Learning, Automated Diagnosis*



MSEC-RS-2025-ECE-005

Advanced Metasurface in Microstrip Antennas for Next-Gen Wifi

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Abstract

This study presents the design and simulation of a metasurface-integrated microstrip antenna for Next-Generation WiFi, 5G, and sub-6G communication systems, specifically targeting n77 and n78 bands. The antenna enhances wireless signal strength by reducing return loss and improving radiation efficiency. It incorporates Split Ring Resonator (SRR) metasurfaces to improve impedance matching and broaden bandwidth. Constructed with Perfect Electric Conductor (PEC) for both the radiating patch and ground plane, it features a microstrip feedline with 50-ohm impedance. A comparative analysis of four substrates—LCP, PTFE, Rogers RO4350B, and Taconic RF-35—was conducted to evaluate performance. PTFE and Rogers demonstrated superior RF characteristics, while LCP offered flexibility for wearable or embedded use. Simulation results confirm the antenna's effectiveness for integration in smartphones, routers, and gaming consoles. Additionally, the design shows potential for scalability to the n79 band and 6G readiness with a PTFE-based prototype.

Keywords: *Metasurface antenna, Microstrip antenna, Split Ring Resonator (SRR), 5G communication, Sub-6G, n77 band, n78 band.*



MSEC-RS-2025-ECE-006

Low Power VLSI Design for IoT and Wearable Devices

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Abstract

Low-power VLSI design has become a critical enabler for next-generation IoT and wearable devices, where stringent constraints on energy consumption, compact form factor, and continuous operation demand highly optimized hardware architectures. This work focuses on design methodologies, circuit techniques, and architectural strategies that significantly reduce power dissipation without compromising system performance and reliability. In IoT sensor nodes and wearable health-monitoring platforms, energy efficiency directly impacts battery lifetime, device portability, and user experience. Therefore, achieving ultra-low power operation through innovative VLSI design approaches is essential for large-scale deployment of intelligent, always-on systems. The abstract presents key low-power design principles such as voltage scaling, subthreshold operation, power gating, clock gating, and adaptive body biasing, which collectively contribute to minimizing switching and leakage power. For IoT edge devices performing lightweight computation and intermittent communication, the adoption of approximate computing, event-driven processing, and memory-centric architectures further enhances energy savings. Similarly, wearables require specialized low-power ASICs and SoCs capable of handling biosignal acquisition, local inference, and wireless connectivity under tight thermal and power budgets. Emerging trends like energy-harvesting circuits, near-threshold computing, and ultra-low-power ML accelerators enable long-term autonomous operation in IoT and wearable applications.

Keywords: *Low-power VLSI, IoT devices, Wearables, Subthreshold design, Power gating, Energy-efficient architectures*



MSEC-RS-2025-ECE-007

Parkinson's Disease - Etiology, Pathophysiology, and Therapeutic Strategies

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Abstract

Parkinson's disease (PD) is the second most common age-related neurodegenerative disorder after Alzheimer's disease, with its prevalence increasing significantly in people over 60. The clinical presentation is highly variable, but cardinal motor features (collectively termed parkinsonism) include resting tremor, muscle rigidity, and bradykinesia (slowness of movement), with postural instability developing in later stages. Non-motor symptoms such as loss of smell (anosmia), sleep disorders (e.g., REM sleep behavior disorder), depression, anxiety, cognitive impairment, and autonomic dysfunction (e.g., constipation, orthostatic hypotension) often precede motor symptoms by years and significantly impact quality of life. The pathophysiology involves the progressive degeneration and death of dopaminergic neurons in the substantia nigra pars compacta, leading to a profound dopamine deficiency in the striatum. A key pathological hallmark is the presence of intracellular aggregates of the protein alpha-synuclein, known as Lewy bodies, although some genetic forms of the disease may not show this pathology. The exact etiology is largely unknown for most cases (idiopathic PD), but it is considered multifactorial, resulting from a complex interplay of genetic susceptibility (about 10% of cases are monogenic) and environmental factors like pesticide exposure and head injuries.

Keywords: *Parkinson's disease, Neurodegeneration, Dopaminergic neuron loss, Substantia nigra, Alpha-synuclein, Lewy bodies, Motor symptoms, Non-motor symptoms, Bradykinesia, Resting tremor, Pathophysiology, Genetic susceptibility, Environmental risk factors, Idiopathic PD, Autonomic dysfunction.*



MSEC-RS-2025-ECE-008

Hybrid Morphology Feature Fusion Framework for Robust Arrhythmia's Classification using ECG Signals

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Abstract

Arrhythmias, being the disturbance in the normal cardiac rhythm, can go unnoticed and lead to consequences such as stroke, heart failure, or sudden cardiac arrest if not caught early. In this study, the focus is on developing a reliable means of arrhythmia detection by combining ECG morphology characteristics with an advanced extraction of signal features. In the MIT-BIH Arrhythmia Database which was annotated clinically, the analysis was performed on five classes of beats, namely, normal, tachycardia, bradycardia, supraventricular, and ventricular ectopic beats. Among many important morphological features pertaining to the shape of P-wave, changes in QRS complex, alterations in T-wave, variations in RR-intervals, and random timing irregularities from beat to beat, further temporal as well as statistical descriptors are combined to describe the morphological characteristics of the ECG segment in much greater detail. The developed system produced excellent results for classification, with sensitivity values of 92.97%, 87.41%, and 94.17%, while the positive predictive values, respectively, of 98.9%, 95.2%, and 90.2% showcased its efficacy of abnormal detection with minimal false positives. These results, however, far exceed the threshold of parametric standards. The results of our study showed that combining ECG morphology and advanced feature extraction greatly enhance arrhythmia detection.

Keywords: *ECG Signal Processing, Arrhythmia Classification, Feature Extraction, Hybrid Feature Fusion, MIT-BIH Arrhythmia Database*



MSEC-RS-2025-ECE-009

Prana – A Mobile Health Care Assistant

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Abstract

Healthcare is evolving through the integration of traditional practices like Siddha medicine and modern technologies such as machine learning. Siddha medicine, an ancient Indian system, offers holistic treatments for ailments like digestive disorders, respiratory diseases, and chronic conditions using herbal and mineral formulations. NGOs enhance access to Siddha healthcare through free medical camps that provide consultations and treatments, benefiting underprivileged communities. Tools like Body Mass Index (BMI) calculators also play a vital role in promoting health awareness by assessing risks of obesity-related diseases such as diabetes and hypertension. Meanwhile, machine learning algorithms like Random Forest, Naïve Bayes, and Decision Tree revolutionize modern healthcare by enabling disease prediction, diagnosis support, and patient classification. By merging Siddha medicine, health awareness tools, and AI-driven approaches, healthcare can achieve personalized, accessible, and efficient solutions, paving the way for a holistic and innovative healthcare framework.

Keywords: *Siddha Healthcare, Machine Learning, Disease Prediction, Holistic Medicine*



MSEC-RS-2025-ECE-010

Blue-Net: Real -Time IoT Tracking and Rescue Alert for Coastal Fishermen

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Abstract

Blue-Net is an innovative IoT-based tracking and safety system designed to ensure the security of fishermen by providing real-time location monitoring and emergency alerts during their fishing trips. In coastal regions, many small-scale fishermen travel up to 10–15 km from the shore, where accidents, engine failures, or adverse weather conditions can pose serious risks. Blue-Net addresses these challenges by integrating GPS tracking, long-range RF communication, and a simple alert system into a compact, cost-effective device suitable for small boats. The system uses a GPS module to capture the exact location of the boat and a 433 MHz long-range RF module to transmit data over distances up to 20 km without relying on mobile network coverage. A central monitoring system receives the data, allowing families or authorities to track the boat's position on a simple interface. An SOS emergency button enables fishermen to immediately send distress alerts with coordinates during critical situations. The device is designed for low power consumption, durability in marine environments, and continuous operation throughout the fishing trip.

Keywords: *IoT, GPS Tracking, Real-Time Monitoring, RF Communication, Emergency Alert Fisherman Safety, Embedded System, Low-Cost Prototype, Geo-Fencing.*



MSEC-RS-2025-ECE-011

Design and Implementation of an FPGA-Based Hardware Verification Accelerator

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Abstract

Verification of digital circuits traditionally relies on software-based simulators such as ModelSim and Vivado Simulator. While accurate, these simulators become extremely slow as the complexity of RTL designs increases, leading to long verification cycles and delayed development. To address this challenge, this project proposes an FPGA-based hardware verification accelerator that executes RTL designs directly on FPGA fabric to significantly reduce verification time. In this work, a selected Design Under Test (DUT)—such as an ALU or a digital processing block—is verified through two parallel approaches: conventional software simulation and hardware-accelerated verification on an FPGA. A synthesizable hardware testbench consisting of a stimulus generator, output checker, and controller FSM is implemented on the FPGA to supply test vectors and validate outputs in real time. The DUT is synthesized and deployed on a Xilinx FPGA platform, enabling execution at hardware clock speeds (tens of MHz), in contrast to the much slower event-driven simulation environment. Performance metrics such as total execution time, number of cycles, throughput, and speedup are measured and compared between the two verification methods. Results demonstrate a substantial improvement in verification speed, highlighting the effectiveness of FPGA-based acceleration. This approach offers a scalable and efficient verification methodology suitable for modern VLSI designs, where verification accounts for a significant portion of development time.

Keywords: *FPGA-based Verification, RTL Simulation, Digital Circuit Verification, Testbench, Xilinx FPGA, ModelSim, Vivado Simulator, LSI Design, Performance Metrics, Stimulus Generator, FSM, Hardware Clock Speeds.*



MSEC-RS-2025-ECE-012

Optimizing Power Allocation in CR-Noma for 5G/6G Cognitive Radio Networks

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Abstract

Cognitive Radio with Non-Orthogonal Multiple Access (CR-NOMA) is emerging as a powerful approach to improve spectrum efficiency in 5G and future 6G networks. Unlike traditional orthogonal access methods, NOMA enables multiple secondary users (SUs) to share the same frequency band at the same time by using power-domain multiplexing. In cognitive radio settings, however, this coexistence requires careful optimization to ensure that primary users (PUs) are protected from interference while SUs achieve reliable and efficient communication. The key challenge lies in power allocation: weaker users need more transmission power to satisfy quality-of-service (QoS) requirements, whereas stronger users can operate with less power and leverage successive interference cancellation (SIC) to decode signals effectively. This optimization problem is inherently non-linear and constrained by power budgets, PU interference thresholds, and SU QoS demands. To address these challenges, researchers have applied a range of methods, from convex optimization for tractable formulations, to metaheuristics like genetic algorithms and particle swarm optimization for more complex cases, and more recently, machine learning techniques such as deep reinforcement learning for adaptive power allocation in dynamic environments.

Keywords: *Cognitive Radio, Non orthogonal Multiple Access, Optimization*



MSEC-RS-2025-ECE-013

Bio Feedback Controlled VFD

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Abstract

The Bio-Feedback Controlled Variable Frequency Drive (VFD) is an innovative automation system designed to adjust motor speed based on real-time human bio-signals. Traditional motor control systems rely on manual adjustment or preset values, which may not respond effectively to the user's physiological condition. This project integrates bio-sensors, a microcontroller, and a VFD to create a smart, adaptive, and automatic motor control system that enhances user comfort, safety, and efficiency. The system begins with a bio-sensors such as a heart rate sensor, muscle activity sensor, or motion sensor detecting physiological changes in the user. These bio-signals represent the user's physical exertion level, stress level, or movement intensity. The detected signal is then transmitted to a microcontroller, which processes and converts it into a meaningful control input. This processed signal is sent to the VFD, which adjusts the frequency of the supply to the motor, thereby regulating its speed automatically. This closed-loop system ensures real-time adaptation of the motor to match the physical state of the user. For example, in a treadmill, if a user's heart rate increases, the motor automatically increases or decreases speed based on predefined safety and performance parameters. The project has significant potential applications in fields such as fitness equipment, patient rehabilitation, assistive medical devices, and smart monitoring systems. Its advantages include energy savings, enhanced safety, improved comfort, reduced manual intervention, and development of a more research-oriented and user-adaptive design. Overall, this project demonstrates how combining bio-feedback with modern motor control technology can provide an efficient, safe, and advanced solution for real-time motor speed regulation.

Keywords: *Bio-feedback, Variable Frequency Drive, Motor Control, Microcontroller, Sensors, Rehabilitation*



MSEC-RS-2025-ECE-014

Low-Cost Edge Computing Telemedicine Kiosk for Intelligent Rural Diagnostics

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Abstract

Rural regions in India continue to face substantial barriers to timely and affordable healthcare due to inadequate infrastructure, long travel distances, limited medical personnel and low levels of digital literacy. Traditional telemedicine solutions partially address these challenges but often rely on smartphones, network stability and trained intermediaries, limiting their deployment in underserved communities. This proposed work presents an AI-powered Telemedicine Kiosk built on a Raspberry Pi-based edge computing platform designed to deliver autonomous diagnostics and real-time teleconsultation without requiring advanced digital skills from users. The system integrates essential biomedical sensors—including HX711 for weight, HC-SR04 for height, MLX90614 for body temperature and MAX30100 for SpO₂—to capture primary health parameters and upload them securely to cloud services for remote clinical access. A multilingual speech-based symptom intake pipeline enables patients to communicate in their native language, wherein voice input is recognized, translated and processed by a large language model (Gemini AI) to extract structured clinical information and identify the appropriate medical specialization. Consultation requests are routed automatically and doctor–patient interactions are facilitated through a WebRTC-based peer-to-peer video communication framework, ensuring low latency and strong privacy even on standard 4G networks. Designed for deployment in village centers and public health posts, the proposed kiosk provides a scalable, cost-effective and field-ready solution to bridge the urban–rural healthcare divide.

Keywords: *Telemedicine, Edge Computing, AI-Assisted Diagnostics, WebRTC, Raspberry Pi, Voice-to-text translation, NLP based symptom extraction, Multilingual Speech Interface*



MSEC-RS-2025-ECE-015

Ragtalk – An Agentic Rag Based Conversational AI

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Abstract

Conversational AI has become an essential tool in many industries, but conventional Retrieval-Augmented Generation (RAG) models face challenges such as poor ranking, limited query expansion, and unstructured response formatting. These limitations reduce accuracy and hinder AI-driven interactions, necessitating more advanced architectures for effective information retrieval. RagTalk, an agentic RAG-based conversational AI, addresses these challenges through a multi-agent framework comprising a Retriever Agent, Ranker Agent, and Response Agent. The Retriever Agent determines whether to extract information from uploaded documents or web searches and retrieves the data, facilitating dynamic query expansion. The Ranker Agent prioritizes retrieved data using similarity-based ranking, ensuring relevance. Finally, the Response Agent enhances clarity by structuring responses with elements such as tables, bullet points, and code snippets. Implemented using Python, Faiss, CrewAI agents, LLM, and Embedding-001, RagTalk significantly improves retrieval accuracy, ranking efficiency, and structured response generation. By setting a new standard for conversational AI, RagTalk offers a scalable and adaptable solution for applications in healthcare, academic research, and customer service.

Keywords: *Conversational AI, RAG, multi-agent framework, query expansion, ranking, structured response, LLMs, CrewAI, Faiss*



MSEC-RS-2025-ECE-016

Intelligent Diagnostics: Advancing Disease Detection with Machine Learning Algorithms

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Abstract

Cardiovascular diseases are among the most serious global health threats because they progress silently and asymptotically until severe complications occur, making early detection essential yet difficult. This silent progression creates the need for diagnostic systems that deliver highly accurate and easily accessible real-time assessments for large populations. However, many existing digital diagnostic platforms fail to meet these demands due to delays, reduced responsiveness, and performance degradation under heavy user loads. To address these shortcomings, this study introduces HealthFaaS, an intelligent and scalable healthcare framework that integrates IoT-based physiological data collection with serverless machine learning for efficient real-time cardiac-risk prediction. Biomedical data captured from IoT devices is securely transmitted to Google Cloud Functions, where machine-learning models run automatically without server management. Five supervised models GM, ANN, XGBoost, SVM, and LightGBM were trained and evaluated using accuracy, precision, recall, F1-score, and AUC, with LightGBM achieving the highest performance at 92.82% accuracy. Scalability tests with varying concurrent user loads and cold-start experiments showed that serverless deployment consistently provides faster response times, higher throughput, better automatic scaling, and shorter initialization delays, especially with increased memory and lightweight runtimes.

Keywords: *Serverless Computing; HealthFaaS; Internet of Things (IoT); Cloud Functions; Heart Disease Prediction; Machine Learning*



MSEC-RS-2025-ECE-017

Leveraging Machine Learning and IoT for Intelligent Crop Recommendation in Modern Agriculture

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Abstract

Agriculture faces numerous obstacles, including resource inefficiency, soil degradation, and climate variability, despite being essential to human survival. The goal of this research is to improve agricultural sustainability and productivity by developing an intelligent crop recommendation system using real-time sensor data. By combining advanced machine learning (ML) models with Internet of Things (IoT) sensors, this research provides farmers with useful information that enables them to enhance crop selection according to specific field conditions. IoT sensors recorded critical information, including soil pH, macronutrients (K, P, N), micronutrients (Zn, S), temperature (T2M_MAX-Su), precipitation (PRECTOTCORR-Su), wind direction (WD10M), and soil moisture (GWETTOP). The collected data underwent preprocessing, which included outlier removal, feature scaling, and missing value imputation. Feature selection techniques were used to identify the most important factors for crop prediction, while marginally relevant characteristics were eliminated to improve model performance. A number of machine learning algorithms were evaluated, including Random Forest (RF), Support Vector Machines (SVM), and Naive Bayes (NB). Hyperparameter adjustment and ensemble learning methods such as gradient boosting machines (GBM) and XG Boost were employed to improve accuracy. A dependable classification model was ensured by addressing data imbalance issues with the Synthetic Minority Over-sampling Technique (SMOTE).

Keywords: *Precision Agriculture, Random Forest, XG Boost, SVM,*



MSEC-RS-2025-ECE-018

A Secure and Low-Power Built-In Self-Test using Adaptive Weighted LFSR and Crypto - MISR

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Abstract

Testing integrated circuits has always involved balancing coverage, cost, and efficiency. As chips become more complex and security issues grow, traditional Built-In Self-Test (BIST) methods face three main challenges: high power use, incomplete fault detection, and risks of test port misuse. This paper offers a BIST framework that combines an Adaptive Weighted Linear Feedback Shift Register (LFSR) with a Crypto-Enhanced Multiple Input Signature Register (MISR) to tackle these problems. The adaptive LFSR creates smarter patterns by biasing certain bits, which improves fault detection while keeping power low. The Crypto-MISR mixes responses with dynamic keys before compacting them, making signatures less predictable. A lightweight FSM controller with clock gating further cuts down on idle power. The design is detailed in Verilog and tested through simulation on an XOR-based circuit. Results show that the architecture generates accurate signatures, reduces switching activity, and enhances security. This approach works well for IoT devices, automotive systems, and medical electronics, where both test reliability and data protection matter.

Keywords: *Built-In Self-Test (BIST), Adaptive Weighted LFSR, Crypto-Enhanced MISR, Fault Detection, Low-Power Testing*



MSEC-RS-2025-ECE-019

Monoaural Speech Enhancement in Multi Speaker Environment

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Abstract

This paper presents a novel Convolutional Neural Network (CNN)-based framework for speech enhancement, which integrates deep learning techniques in multi speaker environment. Throughout the last several decades many researches have proposed several technics to enhance the speech signal captured in environmental noise. The proposed model exploits both spatial and temporal information by analyzing two preceding and two succeeding acoustic frames along with the current frame, enabling it to effectively capture speech dynamics and differentiate speech from background noise. A significant contribution of this work is to enhance the target speaker's speech in a multi speaker environment. Experimental evaluations using standard noisy speech datasets demonstrate that the proposed approach consistently performs better across various signal-to-noise ratio (SNR) levels, confirming its effectiveness in enhancing both perceptual quality and intelligibility of speech.

Keywords: *CNN, Multispeaker, Speech Enhancement, Speech Quality, Speech Intelligibility*



MSEC-RS-2025-ECE-020

Security Improvement of Hight Cryptographic Algorithm

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Abstract

HIGHT (High-performance and Lightweight Cryptography) is a lightweight block cipher designed for environments with limited resources, such as embedded systems and IoT devices. This paper explores the security aspects of the HIGHT algorithm and proposes improvements to enhance its resilience against various attack. we focus on strengthening the key schedule, refining the permutation steps, and introducing advanced techniques. We aim to increase the algorithm's resistance to known attacks, such as meet-in-the-middle and related-key attacks, without significantly impacting its efficiency. This work contributes to the development of more secure cryptographic solutions for resource-constrained applications.

Keywords: *HIGHT, Cryptocore, Cipher, Resource strained applications.*



MSEC-RS-2025-ECE-021

Smart Fertilizer Recommendation System

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Abstract

Agriculture is the backbone of India's economy, and crop productivity highly depends on the balanced use of fertilizers. Traditional fertilizer application is based on generalized guidelines, often ignoring soil health, nutrient needs and weather conditions. This results in soil degradation, lower yield and financial loss to farmers. "The Smart Fertilizer Recommendation System" aims to overcome these issues, offering customized fertilizer suggestion for each crop and soil type. It considers three key factors – soil nutrient composition, crop type and weather. Real time weather data is integrated from the open weather API combined with soil and crop inputs for precise fertilizer selection. Unlike static data sets, the model adopts a dynamic recommendation process, introducing fertilizer variation to avoid dependency on one type. This system is developed using python, stream lit, plot, Rest API's, ensuring interactivity and simple user interface. Farmers or agricultural officers can easily input soil and crop data, and instantly receive fertilizer advise with weather insight. It provides a scalable base for IOT integration and machine learning expansion. Future versions can connect with soil sensors and mobile platforms. Thus the project contributes to smart agriculture sustainability and cost efficiency.

Keywords: *Dynamic Fertilizer Recommendation, Weather API Integration, Stream Lit Application*



SparkSense: Next Level Predictive Maintenance and Monitoring Platform

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Abstract

Unexpected failures of electric drives create costly downtime and safety risks, while traditional reactive or periodic maintenance cannot reliably prevent breakdowns. SparkSense delivers a real-time, IoT-enabled predictive maintenance solution that detects faults before they occur. The system integrates low-cost hardware and edge intelligence. An ESP32 microcontroller acquires temperature, vibration, and current data through a DHT22 sensor, MPU6050 accelerometer, and ACS712 current sensor. Verified using Wokwi simulation, the ESP32 runs custom Arduino firmware (sketch.ino) and communicates alerts via the lightweight MQTT protocol. On the software side, motor-health datasets (classData.csv, detect_dataset.csv) were processed in Google Colab and a compact TinyML anomaly-detection model was trained with TensorFlow. The model was optimized and converted to TensorFlow Lite Micro, enabling on-device inference without constant cloud dependence. When abnormal patterns are detected, the ESP32 sends instant alerts and live sensor readings to a cloud dashboard (e.g., ThingsBoard/AWS IoT) for operator action, enabling proactive maintenance scheduling. By combining machine learning, edge computing, and IoT connectivity, SparkSense provides a reliable, cost-effective platform that enhances the safety and efficiency of industrial electric-drive systems.

Keywords: *IoT, predictive maintenance, anomaly detection, TinyML, edge computing, industrial monitoring*



Smart Traffic Synchronization and Route Booking System for Chennai: A Green-Wave Based Approach to Urban Mobility

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Abstract

Urban congestion is one of the most pressing challenges in modern cities, particularly in India where rapid motorization has outpaced infrastructure expansion. Traditional traffic management relies on static signals and generalized control, which often leads to unnecessary delays, fuel wastage, and commuter stress. This paper proposes an innovative solution which consists of a Smart Traffic Synchronization and Route Booking System. By allowing individuals to input their source and destination through a mobile Application, the system calculates the optimal departure time when traffic density is minimal and coordinates traffic signals along the route to create a personalized “green wave” assuming the commuter maintains an average speed of 40 km/h. The study explores the applicability of this system in Chennai, a city with one of the highest congestion levels in India, particularly during weekday peak hours. Using predictive traffic models, optimization techniques such as time-dependent shortest path (TDSP), MAXBAND, Max-Pressure Control, and Model Predictive Control (MPC), and communication protocols like MQTT and NTCIP, the system integrates predictive analytics, real-time synchronization, and commuter guidance. The proposed system shows potential to reduce travel times by up to 50%, lower fuel consumption, and enhance commuter experience. By merging predictive departure-time planning with coordinated green waves, the system transforms congested urban travel into a smooth and predictable experience.

Keywords: *Smart Traffic Synchronization, Predictive Analytics, Green Wave Control, Route Optimization*



Smart and Sustainable Fast Charging Infrastructure for Electric Vehicles: Photovoltaic Integration, Bidirectional Zeta-KY Converter, and Chaotic Dragonfly Optimized Control

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Abstract

The rapid adoption of electric vehicles (EVs) demands charging systems that are both energy-efficient and environmentally sustainable. This work presents a fast-charging station that integrates photovoltaic (PV) generation, battery energy storage, and grid support into a unified framework. A modified bidirectional Zeta-KY converter is employed to regulate bidirectional power flow, achieving high efficiency in both buck and boost operation. To enhance stability and accuracy, a Chaotic Dragonfly Optimized Proportional–Integral (CDO-PI) controller is implemented, offering rapid dynamic response with minimal steady-state error. A dynamic energy management algorithm coordinates the power contributions of PV, battery, and grid, ensuring reliable charging while reducing dependency on the utility supply. Simulation and experimental results confirm the effectiveness of the proposed design, with the converter reaching efficiencies above 96% and the controller demonstrating fast settling time and zero overshoot. The combined system establishes a sustainable, cost-effective, and reliable solution for future EV charging infrastructure.

Keywords: *Fast charging, Photovoltaic system, Bidirectional Zeta-KY converter, Chaotic Dragonfly Optimization, Power management*



Early Diagnosis of Multiple Sclerosis using Swept- Source Optical Coherence Tomography and Convolutional Neural Network

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Abstract

Multiple Sclerosis (MS) is a neurological chronic disease that impacts the central nervous system by destroying the myelin sheath of nerve fibers. Early identification of MS is critical to decelerate the progression of the disease and allow for early treatment. This project introduces a new method for the early diagnosis of MS based on Swept-Source Optical Coherence Tomography (SS OCT) images and a Convolutional Neural Network (CNN)-based classifier. The process involves the collection of SS-OCT retinal images, preprocessing with contrast and edge-based enhancements, and feature extraction for retinal layer abnormalities. Important diagnostic markers like layer thickness, contrast, and homogeneity were calculated and utilized as input features to train the CNN classifier. This process not only simplifies the diagnosis but also improves accuracy in the detection of early signs of MS-related degeneration in the retina. Our designed CNN model reaches high accuracy for identifying MS-affected retinal structures compared to healthy structures. Experimental results support the system's robustness and generalizability. The work proves the capabilities of AI-aided OCT diagnostics as an inexpensive, non-invasive diagnostic tool for neurological disorder early diagnosis.

Keywords: *Multiple Sclerosis, SS-OCT, CNN Classifier, Retinal Layer Analysis, Early Diagnosis, AI in Healthcare*



A Smart Speak Through Signs System: Translating Gestures into Voice

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Abstract

Communication is a fundamental aspect of human interaction, yet individuals with speech and hearing disabilities often face barriers in expressing themselves effectively. The Gesture-to-Voice Conversion System for Deaf and Mute Individuals aims to bridge this gap by leveraging advanced technologies such as computer vision and machine learning. This smart assistive device is designed to recognize Sign Language gestures and convert them into audible speech, enabling users to communicate effortlessly in various social settings. The system functions by capturing static hand gestures through a vision-based model, processing the input using machine learning algorithms, and outputting the corresponding voice message via an integrated speaker. By providing a seamless and efficient communication platform, this project empowers individuals with speech and hearing impairments to navigate daily interactions independently. The aim of this project is to develop an Assistive Sign Language Communication Device using a Raspberry Pi that enables individuals with speech or hearing impairments to communicate effectively. The system utilizes a USB camera to recognize alphabet signs through a sign recognition Python model and converts them into voice output. Additionally, a keypad with predefined messages allows users to send common phrases as voice outputs for quick communication. With the growing integration of assistive technologies, this system serves as a pioneering step toward inclusive communication and accessibility, improving the quality of life for individuals facing speech and hearing challenges.

Keywords: *Gesture-to-Voice Conversion , Raspberry Pi , sign recognition Python model, Sign Language gestures*



MSEC-RS-2025-EEE-006

Performance Analysis of Voltage Control in Microgrid using Fuzzy Controller

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Abstract

A Micro-Grid is a local energy grid with control capability, which means it can connect in both on-grid as well as off-grid. Based on power generation needs, micro-grid installation with DG technologies are not suitable to direct connection to the grid, for that power electronics converters are needed. A micro-grid can be intended as backup power to the main power grid during periods of heavy demand. The control parts are very responsible in Micro-Grid (MG) for a reliable and stable operation. The main control functions in the micro-grid are voltage, frequency, active power, reactive power. For any operating condition, the voltage and frequency control is important. Hence, without a stable and accurate control the micro-grid implementation is not feasible, thus the control aspects of the micro-grid are the most important. In the earlier research works, various control strategies are proposed and also adding new control techniques. In this work, the voltage control of a micro-grid implementation of islanding mode is modeled with hierarchical control scheme using fuzzy controller in MATLAB Simulink Software is implemented and analyzed the results.

Keywords: *Micro-Grid, Distributed Energy Resources, Distributed generation, Voltage Source Inverter, Voltage Source Converter, Fuzzy Secondary Control*



Automated Trolley System for Disabled and Aged

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Abstract

The rapid expansion of supermarkets and retail industries has increased the demand for smarter shopping solutions that reduce customer effort and enhance convenience. Traditional shopping involves manually searching for products, pushing heavy trolleys, and waiting in long billing queues, which often leads to dissatisfaction. To address these challenges, this project proposes a Smart Trolley System that integrates embedded hardware, wireless communication, and a mobile application to modernize the shopping process. The system is built around a microcontroller that manages Bluetooth communication, motor drivers, and control modules. An Android application, developed in Kotlin, provides two major features: product search with location guidance and trolley control using both voice commands and button inputs. The product search function displays the exact row and zone of items from both front and back gates, allowing customers to navigate quickly within the store. Meanwhile, the trolley can be operated in multiple directions—forward, backward, left, right, and stop—making it user-friendly and efficient for customers of all age groups. In addition, the Smart Trolley incorporates an automated billing system, minimizing human intervention at checkout counters and reducing customer waiting times. Testing and implementation confirm that the system reliably performs product search, responds accurately to user commands, and ensures smooth trolley movement. By combining smart navigation, product assistance, and billing automation, the proposed system significantly enhances the retail experience while contributing to the broader vision of IoT-enabled smart shopping environments.

Keywords: *Smart Trolley System, Embedded Systems, Wireless Communication, Kotlin Programming, Product Localization, Mobile human-machine interface*



Design & Implementation of a Digital Twin Based BMS for SOH Prediction in EV Battery Packs

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Abstract

The rapid rise of electric vehicles has increased the need for intelligent Battery Management Systems (BMS) capable of ensuring safety, extending battery life, and delivering accurate real-time State of Health (SOH) estimation. Conventional BMS methods, relying on empirical models and periodic diagnostics, struggle to provide reliable predictions under dynamic operating conditions. This work proposes a Digital Twin-based BMS (DT-BMS) that integrates physics-based electrochemical models, machine-learning algorithms, and continuous sensor data. The system includes a physical layer with voltage, current, temperature, and impedance sensors, and a digital twin layer using equivalent circuit models, electrochemical impedance spectroscopy (EIS), and adaptive predictive algorithms. A parameter-adaptive lithium-ion model tracks capacity fade and resistance growth, while real-time calibration maintains high-fidelity behavior. LSTM networks and Gaussian Process Regression (GPR) support accurate SOH forecasting, aided by low-latency edge computing. Results show that the DT-BMS achieves SOH prediction errors below 2–3% and enables early detection of degradation trends, improving safety, reliability, and battery life. This makes it a strong candidate for next-generation EV battery management.

Keywords: *Digital Twin, BMS, SOH Prediction, EV Battery, Machine Learning, Electrochemical Modeling*



Proceedings of Third Research Summit 2025

Innovate, Simulate & Solve (RSISS'25)



MSEC-RS-2025-EEE-009

Life Cycle Assessment of EV Batteries

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Abstract

Electric vehicles (EVs) are rapidly transforming transportation, with lithium-ion batteries (LIBs) enabling high energy density and reduced greenhouse gas emissions during operation. However, the sustainability of EV batteries depends on impacts across their entire life cycle, not just their use phase. Life Cycle Assessment (LCA) provides a holistic approach to evaluate environmental effects from raw material extraction to production, use, recycling, and disposal. This study highlights that upstream processes—especially lithium, cobalt, and nickel extraction—carry high carbon footprints, ecological risks, and resource challenges. Although the use phase has comparatively lower impacts, it is strongly affected by grid electricity, charging behavior, and vehicle lifetime. End-of-life management offers major opportunities for impact reduction through second-life applications and recycling. Second-life batteries used in stationary storage extend service life and reduce waste. Recycling technologies such as hydrometallurgical, pyrometallurgical, and emerging direct recycling have varying recovery efficiencies and environmental trade-offs. LCA comparisons show that circular strategies like reuse, eco-design, and closed-loop recycling can reduce lifecycle emissions by 30–40%.

Keywords: *Lithium-ion batteries (LIBs), Life Cycle Assessment (LCA), Sustainable recycling, Second-life applications, Hydrometallurgical recycling, Pyrometallurgical recycling, Direct recycling, Circular economy, End-of-life (EoL) management.*



MSEC-RS-2025-EEE-010

RENEGA: Real-Time Monitoring of Biogas Slurry in Rural Plants using GSM and Piezoelectric Energy Harvesters

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Abstract

India has been a major producer of biogas for nearly a century, with the first plant built in Mumbai in 1859. Today, more than five million biogas plants—from household units to large commercial facilities—convert organic waste into clean energy. States such as Maharashtra, Tamil Nadu, and Uttar Pradesh are leading this growth, supporting India's goal of achieving 200 GW of renewable energy by 2030. National programs like SATAT and NNBOMP have further accelerated biogas adoption, especially in rural areas. Despite this progress, biogas plants face challenges in monitoring the slurry inside digesters. Harvesting energy from slurry vibrations using piezoelectric materials is promising but limited by low and irregular output. To address this, the project proposes using highly sensitive piezoelectric modules from Mide Technology and combining them with solar panels to create a stable hybrid energy source for continuous device operation. Monitoring parameters such as temperature, pH, and slurry level is difficult due to noise, harsh environments, and weak sensor signals. This work integrates advanced signal-conditioning circuits and low-power microcontroller algorithms that filter noise, auto-calibrate data, and deliver accurate, reliable measurements for operators. Communication issues in remote rural areas are tackled through a dual system: GSM for regular connectivity and LoRaWAN for long-range, low-power backup communication.

Keywords: *Biogas slurry, GSM, MQTT, Piezoelectric energy harvesting, Anaerobic digestion, Bokashi System*



MSEC-RS-2025-EEE-011

Design of a Single-Switch Converter with High Step-Up Gain and Low Diode Voltage Stress

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Abstract

This study validates the presence of a high step-up converter capable of delivering significantly improved voltage gain while maintaining minimal switching stress on the semiconductor devices. The proposed converter achieves its enhanced performance by operating at a relatively small duty ratio and employing a low turns ratio in the coupled inductor, which together contribute to achieving a high step-up conversion ratio without sacrificing efficiency or component reliability. The coupled inductor plays a crucial role by storing magnetic energy effectively while simultaneously reducing the impact of leakage inductance, which is a common source of voltage spikes and efficiency losses in conventional converters. By carefully selecting the appropriate turns ratio for the coupled inductor, the voltage stress imposed on the main power MOSFET is substantially minimized, thereby improving device lifespan and reducing the need for high-voltage components. To further enhance performance, a snubber circuit is integrated into the design to suppress voltage spikes generated during switching events. This snubber network reduces power dissipation in the main switch, slows down the rate of rise of voltage and current (dv/dt and di/dt), and limits the peak currents and voltages experienced during turn-on and turn-off transitions. As a result, the switching process becomes smoother and more controlled, significantly improving converter reliability. Additionally, the inductive impedance that remains in series with the main winding is efficiently recycled, contributing to an overall increase in conversion efficiency.

Keywords: *Shifting Stress, Voltage Gain, Snubber Circuit, Voltage Surge And Conversion Efficiency*



MSEC-RS-2025-IT-001

Emotionally Intelligent Humanai Conversations Through Multimodal LLM- Based Emotion Detection and Context-Aware Response Generation

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Abstract

Emotionally intelligent communication is still a big challenge in Current conversational AI systems, as most generate generic responses without understanding the user's emotional state. This limitation often leads to interactions That sounds mechanical, insensitive, and disengaging. The following project covers the work of in particular, the following sections focus on how these issues can be addressed through the development of a multimodal emotion-aware conversational framework. that integrates facial emotion recognition with Large Language Model-based Contextual understanding: The leading aim is to devise an adaptive system where Real-time facial emotion detection and textual intent analysis work in tandem to provide Empathetic responses and context-sensitive applications can only serve to enhance a person's Engagement, emotionally appropriate communication, and the bridging of gaps. Its scope is limited between the understanding of human and machine. The proposed framework is likely to enhance conversational relevance, emotional sensitivity, and user satisfaction form the basis for Future developments in emotionally adaptive AI systems.

Keywords: *Emotionally Intelligent Communication, Multimodal Emotion Recognition, Facial Emotion Detection, Large Language Models, Adaptive Conversational AI*



MSEC-RS-2025-IT-002

Multi-Modal Symptom Analysis and Risk Prediction using Pretrained AI Models in Telemedicine

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Abstract

In modern medical practice, concise summaries of patient questions are essential for accurate and efficient care. However, most existing methods focus solely on text, overlooking the value of integrating visual data. To address this gap, this research introduces a multimodal system that merges textual queries with medical images to enhance the extraction of clinically relevant information. The proposed system starts with a gradual fusion decoder BERT with ViT to produce high-resolution feature maps and predict disease categories with precision. To enrich the contextualization of medical queries, it applies a Multi-Agent Contextualized Diffusion Model (MACDM) that integrates cross-modal information to produce a better summary. While BioClinicalBERT classifies severity into Low, Medium, or High. For low and medium cases, self-care remedies are suggested using the MediSearch API; for high severity, the system flags the case for medical intervention. This solution improves accessibility to timely and accurate healthcare, particularly in underserved regions, by automating risk prediction and enabling multilingual, AI-assisted self-care recommendations.

Keywords: *Artificial intelligence, Gradual fusion decoder, large language model, Medical question summarization, Healthcare, Vision transformer, Gradual fusion decoder, BioClinicalBERT, Telemedicine*



MSEC-RS-2025-IT-003

Knowledge Graph Construction for Agricultural Decision Support using Deep Learning

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Abstract

The work Knowledge Graph Construction for Agricultural Decision Support using Deep Learning project aims to transform unstructured agricultural data into a structured, query able knowledge graph that provides actionable insights for farmers and researchers. Traditional agricultural information is fragmented; thus, it becomes challenging to extract relationships between important entities like crops, soil types, weather patterns, pests, and diseases. To address this issue, the system automatically collects data from research papers, web articles, and real-time APIs to build a comprehensive and constantly updating dataset. The text is preprocessed and analyzed by a fine-tuned RoBERTa model for Named Entity Recognition (NER) to extract key agricultural entities, while relationships between these entities are extracted using a BERT-CRF model. The extracted knowledge is then stored as a knowledge base in Turtle (.ttl) format which is then used with NetworkX to construct a knowledge graph. In addition, the user may upload their own agricultural texts to develop specific graphs, thereby improving accessibility and interactivity. The knowledge graph will allow complicated relationships in agricultural data to be looked up and visualized, allowing decisions to be made on crop selection, soil management and pest management.

Keywords: *Knowledge Graph Construction, Agricultural Decision Support, Deep Learning in Agriculture*



MSEC-RS-2025-IT-004

Multi-Agent LLM Framework for Autonomous Syllabus Generation and Validation in University Curriculum Design

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Abstract

The rapid adoption of Large Language Models (LLMs) within the education sector has improved processes such as assessment generation, Bloom's taxonomy mapping, and draft curriculum writing. Nevertheless, their present approaches are fragmented, inadequate, lacking in policy compliance, unable to scale, and do not meaningfully integrate stakeholders. This research focuses on developing a multi-agent LLM framework capable of autonomously generating and validating university syllabi across disciplines, ensuring that these syllabi align with all relevant guidelines, including those of the UGC, AICTE, and NEP, incorporate current industry trends, and consider feedback from faculty and accrediting bodies. The framework employs specialized agents that interpret course requirements, generate structured syllabi, validate policy alignment, enrich content with interdisciplinary insights, and iteratively refine outputs through feedback.

Keywords: *Large Language Models (LLMs), Multi-Agent Framework, Automated Syllabus Generation, Policy Compliance (UGC/AICTE/NEP), Curriculum Design, Higher Education Innovation.*

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ISBN: 978-93-94412-32-3



MSEC-RS-2025-IT-005

AI-Driven Modelling of Grit for Predicting Employee Retention and Long-Term Success in Recruitment Systems

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Abstract

Common recruiting processes value cognitive ability, technical skills, and experience more, whereas less attention is placed on non-cognitive factors such as perseverance or dedication. In this work, we have AI to assess Grit – a trait of sustained interest and effort toward long-term achievement – in job candidates in many cases. By using NLP, Machine Learning, and psychometric validation, the system looks for proxy indicators of Grit in resumes, behavioural interviews, and digital communications. The model will be constructed and cross-validated using Duckworth's Grit Scale and HR performance data on retention prediction, as well as job performance. The end product becomes an interpretable, ethical, and bias-free AI framework for measuring Grit in the hiring space.

Keywords: *Grit, AI in Recruitment, Machine Learning, NLP, Employee Retention, Psychometrics, HR Analytics*



MSEC-RS-2025-IT-006

Predicting Onset Dialysis for CKD Patients using BERT & LSTM

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Abstract

Chronic Kidney Disease can progress silently to End-Stage Renal Disease (ESRD), where dialysis becomes essential, making early prediction information important for timely treatment. Traditional scores of risk, however, lack the capture of longitudinal patterns in labs, demographics, comorbidities and notes, are limited by imbalance and poor interpretability. As such, this project proposes an advanced decision support system to predict the onset of dialysis within 6–12 months and enable risk stratification. A hybrid deep learning model it combines LSTM for sequential lab data with BERT-based transformers for clinical text, supported by a multimodal framework that integrates structured and unstructured EHR data with explainable AI SHAP, with transparency assured, along with earned trust, this approach provides the robust and interpretable solution to improve early CKD management and dialysis prediction with conventional risk category and other possibilities disease based on this data.

Keywords: *Chronic Kidney Disease, LSTM For Sequential Lab Data, BERT Based Transformers, Prediction 6-12 Months*



MSEC-RS-2025-IT-007

LLM-Based Multi-Agent System for Agricultural Residue Utilization Planning

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Abstract

Efficient crop residue management is a major challenge, as large amounts of biomass are still burned due to a lack of clear utilization strategies. This causes air pollution, soil degradation, and loss of valuable resources. This project introduces an LLM-based multi-agent planning system that analyzes residue type, quantity, soil data, and regional constraints to generate optimized utilization plans. A Neo4j knowledge graph grounds the LLM with verified information, reducing hallucination and improving reliability. Users review and refine outputs before receiving an explainable plan covering composting, biochar, biogas, and livestock feed. Built with Python, Flask, Neo4j, and OpenRouter APIs, the system enables transparent, data-driven decision-making.

Keywords: *LLM, Flask, Neo4j, APIs*



MSEC-RS-2025-IT-008

AI-Driven Legal Risk Evaluator for Startup Compliance Management

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Abstract

Startups often struggle to interpret complex legal and regulatory obligations, resulting in compliance gaps during their early stages of development, which can lead to accidental legal exposure. The AI-Driven Legal Risk Evaluator solves this problem by utilizing sophisticated Natural Language Processing (NLP) to comprehensively analyse startup ideas and business plans to identify risks across key areas, including data privacy, intellectual property, licensing, and AI ethics. A rule-based compliance engine links each identified risk area to applicable legal frameworks, such as GDPR, HIPAA, and the IT Act. Using the Scikit-learn framework, a quantitative risk score is calculated, and severity is assessed. The evaluator also contains a LIME-based explainable module for clear justification of the prediction. Overall, the AI-Driven Legal Risk Evaluator provides fast, actionable, and transparent legal insights into startup ideas and can help founders proactively address legal compliance issues, especially if assessed to be high risk, with each risk category offering specific mitigation steps designed to streamline early compliance and reduce dependency on expensive legal consultations.

Keywords: *Legal Risk Analysis, NLP, Compliance Automation, Risk Scoring, Explainable AI, Legal Framework Mapping*



MSEC-RS-2025-IT-009

AI-Powered Bidirectional Sign Language Translation System

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Abstract

A two-way sign-language translation system is proposed to offer an improved communication environment for hearing-impaired and non-hearing-impaired individuals. This system will use a combination of CNNs, together with Bi-LSTM networks, to represent both the spatial and temporal features of the signs. It will have real-time processing along with context-based translation to ensure accuracy and natural interaction. The key features of the proposed system are live gesture recognition, efficient text generation, and the possibility of support in several languages. Further enhancement will be made towards expanded language coverage, superior accuracy in gesture recognition, and addition of voice output. This will give the user an easier, more effective, and friendlier communication experience.

Keywords: *Sign Language Translation, CNN–BiLSTM, Real-Time Gesture Recognition, Inclusive Communication*



MSEC-RS-2025-IT-010

AI-Based Illegal Fishing and Zone Violation Alert System using GRU

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Abstract

The choice of an AI-Based Illegal Fishing and Zone Violation Alert System aims to prevent fishermen from encroaching on restricted or international waters by accident. The input to the system is the real-time GPS data. The data is first filtered through a Kalman Filter to improve the position accuracy and eliminate noise. The next step employs a Gated Recurrent Unit (GRU) model to predict the boat's future location to ascertain whether it is nearing a restricted area. The predicted locations and the actual enforcements are continually verified by geofencing checking the safety of the area. Should any vessel come close to a restricted zone, voice alarm activation becomes immediate. The vessel alert facilitates fishermen's awareness of their current state. By improving maritime safety, this intelligent AI solution would thus help in taking care of unintended boundary violations while enforcing ethical and legal fishing practices.

Keywords: *GRU-based Trajectory Prediction, Vessel Movement Forecasting, Kalman Filter Noise Reduction, Predictive Modeling for Fisheries, AIS/GPS Data Analytics*



MSEC-RS-2025-IT-011

Lime-Based Explainable Framework for Pneumonia Detection

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Abstract

Medical image diagnosis by deep learning algorithms like Convolutional Neural Networks (CNNs) has proved to be highly accurate in identifying abnormalities from chest X-rays. Yet these models tend to act as black boxes, giving predictions without good reasons, which restricts their acceptance amongst clinicians. In addressing this problem, the system proposed here, Seamless Workflow Integration of Explainable AI (XAI) Systems, integrates CNN with Local Interpretable Model-Agnostic Explanations (LIME) for increased transparency and trust in diagnostic results. Not only does the system classify chest X-rays as Normal, Pneumonia, or Tuberculosis but also outputs visual heatmaps where the relevant parts of the image that contributed to the model's choices are indicated. This explainable output closes the trust gap between clinicians and AI by offering transparent, visual reasoning behind every diagnosis. Additionally, the system merges naturally with current Electronic Health Record (EHR) workflows to support decisions in real-time and contextually. By encouraging explainability, ethical deployment, and clinician trust, the suggested framework is a big leap toward reliable and transparent AI deployment in healthcare diagnostics.

Keywords: *Explainable AI, LIME, Pneumonia, CNN, Chest X-ray, Medical Imaging, X-rays.*



MSEC-RS-2025-IT-012

LLM-Based Regulatory Compliance Bot for Corporate Governance

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Abstract

Compliance teams at corporate organizations rarely have the time to keep internal policies aligned with constantly changing regulations and standards of Environmental, Social, and Governance (ESG). In this project, we introduce an AI-specific Regulatory Compliance Bot that uses Streamlit and Google Gemini to provide automated comparison and analysis of documents. Users upload documents and the regulatory process extracts, cleans, and structures the text at the level of clauses to identify conflicts, gaps, and mismatches between the different documents. Further, the bot generates a compliance checklist that is standardized and should improve accuracy, review time, and visibility. A key feature of this project employs breaking documents into additional smaller nuggets of clauses paired with a smart fallback logic engine which will provide clean, consistent, and traceable outputs. This will improve governance workflows and decision support for organizations. The bot can also allow for quick queries to be made in an interactive reporter to check a clause. Lastly, the bot reduces the need for manual legal reviews for documentation to remain compliant with rapidly changing norms and regulations.

Keywords: *Regulatory Compliance, Google Gemini, ESG, AI Automation*



MSEC-RS-2025-MECH-001

Experimental Investigation on The Mechanical and Corrosion Behaviour of Al/B₄C/TiB₂ Composites Fabricated by the Stir Casting Method

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Abstract

This study investigates the mechanical and corrosion behaviour of aluminum-based Al/B₄C/TiB₂ hybrid composites fabricated using the stir vacuum casting method. Three reinforcement combinations- 3wt.%B₄C/5wt.%TiB₂, 6wt.% B₄C /5wt.%TiB₂, and 8wt.% B₄C/ 5wt.% TiB₂- were introduced into the aluminum matrix to examine the influence of increasing ceramic content on composite performance. Scanning Electron Microscopy (SEM) was used to evaluate the dispersion and morphology of the reinforcement particles, providing insight into their distribution and bonding within the matrix. The results indicated that increasing B₄C content contributed to improved hardness and tensile performance due to enhanced load transfer and matrix reinforcement. Corrosion resistance was evaluated in a 3.5% NaCl solution to simulate saline conditions, revealing the composites' behavior under chloride exposure. X-ray Diffraction (XRD) analysis confirmed the presence and stability of B₄C and TiB₂ phases and helped identify any additional phases formed during solidification. Overall, the incorporation of B₄C and TiB₂ significantly enhanced both the mechanical and corrosion characteristics of the aluminum composites, making them suitable for lightweight engineering applications.

Keywords: *Mg alloys, composites, vacuum casting*



MSEC-RS-2025-MECH-002

Performance Comparison of Analytical and Experimental Work of Nanofluid Based Parabolic Trough Solar Collector

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Abstract

Among various solar energy harvesting technologies, the parabolic trough solar collector (PTSC) remains one of the most reliable and widely implemented systems for domestic and industrial thermal applications. In the present work, Al₂O₃–deionized water nanofluids were synthesized at volume concentrations of 0%, 0.5%, 1.5%, 2.5%, and 3.5% and tested in a PTSC at mass flow rates ranging from 0.02 to 0.06 kg/s during the period from 8:00 a.m. to 4:00 p.m., establishing a strong experimental foundation. Building upon these results, an analytical investigation was conducted to evaluate the influence of a 5% rise in ambient temperature on PTSC performance, without altering any other system properties or operating parameters. The results indicate a slight enhancement in performance parameters under elevated ambient temperature conditions.

Keywords: *Parabolic trough solar collector, Nanofluid, Ambient temperature, Influencing factors, Al₂O₃*



MSEC-RS-2025-MECH-003

Analysis of Stirrer Speed and Enriched Dielectric on the Surface Integrity of Inconel 718 During EDM with a Copper Electrode

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Abstract

The research on Perovskite based photovoltaic is rapidly increasing in renewable energy, solar cells fields because its efficiency reached above 25.5%. Although, the number of efforts has been made to increase the lifetime and stability, still several complicated processes are required to achieve better structural and chemical stability. In this paper, perovskite material is prepared in different molar ratio combination of PbI₂ (lead Iodide) and MAI (Methyl Ammonium Iodide) for better inter diffusion and structural ordering. The Synchrotron source-based X-ray Diffraction measurements were performed in order to optimize, understand the structural properties of such films. The synchrotron source based structural study of perovskite material paves a way to understand diffused multi-layers (ITO/PbI₂/MAI) at different angle of incidence and further improvement of such films.

Keywords: *Perovskite Materials, Synchrotron source-based X-ray, Inter Diffusion.*



MSEC-RS-2025-MECH-004

Tribological Behavior of Titanium-Coated Aluminum Composites Under CNC Machining Conditions

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Abstract

This study investigates the tribological performance of titanium-coated aluminum composites when subjected to CNC machining environments. The application of a thin titanium coating is expected to enhance surface hardness, reduce friction, and improve wear resistance during high-speed machining. Experimental trials were conducted using standardized CNC turning and milling operations under varying cutting speeds, feed rates, and tool-work piece interaction conditions. The coated composite surfaces were analyzed for friction coefficient, tool wear, material removal characteristics, and heat generation patterns. Comparative evaluation with uncoated aluminum composites revealed significant improvements in wear resistance, reduced adhesion tendency, and more stable cutting forces due to the presence of the titanium layer. The results demonstrate that titanium coating serves as an effective surface modification strategy to enhance machining performance and prolong tool life in precision manufacturing applications. This work provides valuable insights for optimizing advanced lightweight materials in high-efficiency CNC machining processes.

Keywords: *Titanium coating; Aluminum composites; Tribology; CNC machining; Wear resistance; Friction coefficient;*



MSEC-RS-2025-MECH-005

Experimental Evaluation of Dynamic Environment During Launch Vehicle Lift-Off

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Abstract

The replacement of the existing liquid-propellant core motor with a semi-cryogenic engine marks a major technological upgrade for the Launch Vehicle Mark-III (LVM3). The semi-cryogenic stage, which uses ISROCENE as fuel and Liquid Oxygen (LOX) as the oxidizer, is designed to significantly enhance LVM3's payload-lifting capability. ISROCENE—an advanced, cleaner hydrocarbon fuel—offers higher efficiency, improved cooling characteristics, and greater thrust potential compared to conventional earth-storable propellants. To support this transition, three new umbilical structures are being incorporated into the existing launch pad infrastructure. These umbilical's will manage critical functions, including propellant loading, electrical connections, and pneumatic services necessary for the semi-cryogenic stage. Before full-scale integration, extensive testing is essential to evaluate the location, behavior, and load responses of these umbilical structures under realistic operating conditions. For this purpose, a vertical hot-flow test was conducted at the AATF/SMP-ETF facility at SDSC-SHAR. Because conducting such tests at full launch-vehicle scale is not always feasible, engineers used a scaled-down model of the system. The spacecraft and associated structural parameters were reduced in a ratio of 1:12.8, allowing accurate simulation of fluid dynamics, thermal loads, vibration response, and plume interaction effects while maintaining manageable test conditions.

Keywords: *Semi-cryogenic engine, ISROCENE fuel, Liquid Oxygen (LOX), Umbilical Structures, Hot flow test, LVM3 (Launch Vehicle Mark-III).*



MSEC-RS-2025-MECH-006

Characterization of Alumina Foam Filter for Engine Exhaust Analysis

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Abstract:

Increasing concern over vehicle emissions has accelerated the need for efficient exhaust filtration materials capable of supporting sustainable automotive technologies. Alumina foam filters have gained attention due to their exceptionally high porosity (80–90%) and ability to withstand temperatures up to 1,600°C. Their open-cell structure not only enhances gas–solid interaction for particulate removal but also ensures smooth gas flow with minimal backpressure. These characteristics make alumina foam an attractive option for reducing particulate emissions such as soot and unburned hydrocarbons while maintaining stable engine performance. This research focuses on the characterization of alumina foam filters for engine exhaust applications, emphasizing their structural, mechanical, and thermal behaviour. SEM analysis revealed a uniform pore size distribution of 20–45 pores per inch (PPI), suitable for efficient filtration. Mechanical testing indicated a compressive strength of 2.8–3.5 MPa, confirming the filter's durability under exhaust-induced stress. Additionally, thermal shock resistance tests showed less than 4% mass loss after repeated cycles between 300°C and 1,200°C, demonstrating the material's ability to endure rapid temperature fluctuations commonly encountered in engine systems. Although a small increase in backpressure (3–5 kPa) occurred, it remained within acceptable limits. When integrated with a catalytic converter, the filter also improved NO_x conversion efficiency by 8–10%. These findings highlight the effectiveness of alumina foam filters in enhancing emission control and supporting cleaner, more efficient engine operation.

Keywords: *Alumina foam filter, Engine exhaust, backpressure, Emission control, Porosity, Thermal stability*



MSEC-RS-2025-MECH-007

Effect of Ultrasonic Cavitation on Squeeze Casting of AA6061 for Fatigue Strength

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Abstract

Squeeze casting is a hybrid metal processing technique that combines the advantages of both casting and forging in one operation. Net shape (or) near net shape components can be manufactured through squeeze casting process. Squeeze casting process is suited for all melting ranges of metals and alloys ranging from lead to iron. But nowadays, lightweight materials like aluminum and magnesium are mostly used in the industries. Aluminum alloy 6061 is one such futuristic material that is widely used to produce automotive and aerospace components. Squeeze casting is a process in which the pressure is applied over the molten metal until the end of solidification. The applied pressure induces the grain refinement that results in enhancement of mechanical properties of the squeeze cast components. Further the introduction of ultrasonic cavitation into the melt induces dispersion and degassing action and it also favors the grain refinement. In this attempt ultrasonic cavitations are introduced while melting AA6061 and the cylindrical components are produced through squeeze casting process. The process parameters chosen (squeeze pressure, die-pre heating temperature, pressure applying duration and ultrasonic cavitation) in a squeeze casting process generates the highest mechanical properties attainable in a cast product. In this project, cylindrical components of AA 6061 are produced using squeeze casting process and their number of cycle to failure are found out.

Keywords: *Al 6061 Alloys, Squeeze Casting, Ultrasonic Cavitation, Taguchi Method, Genetic Algorithm Approaches*



MSEC-RS-2025-MECH-008

Mechanical and Water Absorption Behaviour of Palm Seed Particles Based Hybrid Bio-Composites

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Abstract

In the recent decades, many scholars and researchers are interested in natural fibres today because they are inexpensive, widely available, and easy to process compared to conventional fibres. Due to environmental awareness and growing concern over the greenhouse effect, the construction, automotive, and packaging industries are also looking for eco-friendly materials that can replace conventional synthetic polymeric fibres. Alpaca fiber (AF), palm seed filler (PSF), and polypropylene (PP) were combined in this investigation to create composites that outperformed traditional polymer composites in terms of properties. The alpaca fiber/palm seed filler polypropylene (PP) composites were prepared using a conventional injection molding technique with fiber loadings of 35, 30, 25, 20, 15, and 10% and filler loadings of 15, 10, and 5%. (0, 5, 10, 15, 20 and 25 wt. percent). Polypropylene (PP)/Alpaca fiber/palm seed composites with a filler content of 20 wt. % were found to be more promising than PP/Alpaca fibre composites. Keywords: Hardness, flexural strength, alpaca fiber, Natural fiber, Water Absorption.

Keywords: *Hardness, Flexural Strength, Alpaca Fiber, Natural Fiber, Water Absorption*



MSEC-RS-2025-MECH-009

Microstructure and Mechanical Properties of Al/B₄C/SiC Composites using Stir Casting Method

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Abstract

This study investigates the microstructure and mechanical properties of aluminum matrix composites (AMCs) reinforced with Boron Carbide (B₄C) and Silicon Carbide (SiC) particles, fabricated using the stir casting method. Aluminum was chosen as the base metal due to its lightweight and good ductility, while B₄C and SiC were selected for their superior hardness, wear resistance, and thermal stability. The reinforcements were varied in weight percentages to study their influence on the overall composite behavior. Stir casting, a cost-effective liquid-state fabrication technique, was employed to achieve uniform dispersion of the reinforcements within the aluminum matrix. Microstructural analysis using optical microscopy and scanning electron microscopy (SEM) revealed relatively uniform distribution of B₄C and SiC particles, though minor agglomeration and porosity were observed at higher reinforcement levels. Mechanical testing indicated significant improvements in hardness, tensile strength, and wear resistance of the hybrid composites compared to pure aluminum. However, a reduction in ductility was noted with increasing reinforcement content. The study concludes that Al/B₄C/SiC composites produced by stir casting exhibit a favorable balance of mechanical properties, making them suitable for applications requiring lightweight materials with enhanced strength and wear resistance.

Keywords: *Metal Matrix composites, Stir casting*



MSEC-RS-2025-MECH-010

Fabrication and Characteristic of Boron Carbide Particulate Reinforced Aluminum Metal Matrix Composites

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Abstract

Aluminium matrix composites (AMCs) play a pivotal role as advanced engineering materials due to their distinct mechanical properties like lightweight, strength, wear resistance, toughness, stiffness...etc. the focus of the work is on fabricating the aluminium (AA6061) matrix composites reinforced with the varied propositions of the percentage of the weight of B_4C particle of size at $220\mu m$ such as 3%, 6% and 9% by means of the stir casting method. The enhancement in the wettability of B_4C particles in the matrix by adding K_2TiF_6 flux into the molten metal, has been significant. Besides, the microstructure and mechanical properties of the fabricated AMCs have also been analyzed. Uniform distribution of the presence of the B_4C particle in the matrix has been confirmed using the Scanning Electron Microscope (SEM) images. It has been found that the tensile strength and hardness of the fabricated AMCs increases phenomenally with the increased content of the B_4C particle.

Keywords: *Metal Matrix Composites, Stir casting, Microstructure, tensile test, Hardness*



MSEC-RS-2025-MECH-011

Experimental Evaluation of Inter Laminate Shear Stress of Cajral Type Fiber Metal Laminates Prepared with Surface Treated Aluminum Sheet by Sobbing Technique

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Abstract

Industries increasingly require novel lightweight materials, and composites are gaining importance due to their superior mechanical performance compared to aluminium alloys, especially in aerospace applications. Fibre Metal Laminates (FMLs), such as the CAJRAL type made of aluminium, carbon, jute, and epoxy, merge the advantages of fibre composites—high specific strength, stiffness, corrosion and fatigue resistance—with the ductility and durability of metals. This work focuses on developing CAJRAL FMLs using the manual hand lay-up method, with aluminium sheets surface-treated using the sobbing technique. Two laminate sequences were fabricated: Ca 0°/Ca 45°/Al/Ju 45° and Ca 0°/Al/Ca 0°/Al/Ju 0°. Their structural behaviour was assessed through ASTM D2344 three-point bending tests. Results show that carbon composite and aluminium layers largely govern damage behaviour in the bending zone, reflecting the complex degradation of hybrid laminates. Fibre orientation strongly influences laminate performance, and high composite–metal adhesion is crucial for maintaining structural integrity. The CAJRAL laminates with 8 and 10 layers achieved bending loads of 376.07 N and 224.91 N, respectively.

Keywords: *Fiber metal laminates, 3 Point bending test, Inter Laminate Shear Stress, Laminate sequencing*



MSEC-RS-2025-TAMIL-001

ததொழில் நுட்பத்ததப் பயன்படுத்தி தமிழ் இலக்கியங்கதை கற்பிக்கும் உத்திதை

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மீனாட்சி சுந்தரராஜன் ப ாறியியல் கல்லூரி, பென்றன -24,
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ஆய்வுச் சுருக்கம்

காலமாண்ைத்திண்ககண் வும் மாணவர்கள்களின் அறிவுத்திணுக்கும் அவர்களிடம் காண் ுடும் தகவல் பதாழில்நுட் அறிவிண்கு ஏண் வும் 21-ஆம் நுண்ைாண்டுத் திண்கறளக பகாண்டு கண்பித்தல் நிகழ்கிைF. ஆங்கில ஆதிக்கம் மிகுந்திருக்கும் இன்றைய பமாழிண் சூழலில் தாய்பமாழிறயண் சுறவ ட கண்பிக்க ுடல புதிய அணுகுமுறகறள ுட ுன்றிய ஆய்வுகளும், ரிந்றரகளும் நித்தமும் நிகழ்ந் பகாண்டிருக்கிணை. க ஸ்புக், யுடியு ு, ஸ்றக ு, லிங்கடுஇன், கேங்கவுட்ஸ், கவர்டு பிரஸ், டம் ளர் க ான்றைவ தமிழ் இலக்கியங்கறள கைபிக்கெ் சிண்ந்த அடி ு றடத் தளமாக விளங்குகிைF. இவைறைபயல்லாம் யன் டுத்திண் தமிழ் இலக்கியங்கறளக கைபித்தலின் வழி மாணவர்களின் கைல் அறடவு உயருகிணைF என் தறன இக்கட்டுறர பதளிவு டுத்FகிணைF.

திறவுத்தசொல்: பதாழில் நுட் க் கருவிகள், தமிழ் இலக்கியங்கள், கைபித்தலில் புதிய அணுகுமுறைகள்.



MSEC-RS-2025-MATH-001

Octree Neutrosophic Sets using Rough Matrix

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Abstract

The octree is a hierarchical tree data structure commonly employed in computer graphics, spatial indexing, and three-dimensional data representation. Traditional octree methodologies generally categorise regions as black (completely shaded), white (unshaded), or grey (partially shaded), aligning with a binary (0–1) logic, with the intermediate state managed in an ad hoc manner. This article extends binary logic to a three-valued logic octree, facilitating a more nuanced and systematic depiction of spatial uncertainty. We present a neutrosophic quadtree framework that extends the traditional quadtree by explicitly representing the components of truth, indeterminacy, and falsity for each node, in accordance with neutrosophic set theory. We propose a consolidated methodology for expressing imprecise spatial data by integrating these two approaches, wherein each node conveys either three-valued (true, false, intermediate) or neutrosophic (T, I, F) information. Additionally, we illustrate how both structures can be represented and examined using rough matrices, employing the concepts of lower, higher, and boundary approximations to measure certainty and indeterminacy inside hierarchical divisions. This comprehensive perspective amplifies the descriptive capability of tree-based spatial models and paves the way for advancements in image reconstruction, feature extraction, and uncertainty management in multidimensional datasets.

Keyword: *Neutrosophic Quadtree, Three-Valued Logic, Spatial Uncertainty, Rough Matrices, Imprecise Spatial Data, Tree-Based Models.*



MSEC-RS-2025-MATH-002

Application of Higher Mathematics in Different Disciplines

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Abstract

This article is based on higher mathematics thoughts and their disciplines. Applied mathematics is a combination of mathematical science and specialized knowledge. Mathematics is applied in various fields i.e., mathematical concepts and procedures are used to solve problems in science, engineering, economics. At present every branch of mathematics has a potential for applicability in other fields of mathematics and other disciplines. The 21st Century mathematics thinking is more to strengthen efforts to bridge the division lines within mathematics, to open up more for other disciplines and to foster the line of inter-discipline research.

Keywords: *Daily life, medicine, general life, Physics, Chemistry, Computer .*



MSEC-RS-2025-MATH-003

Fixed Point Theory

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Abstract

Fixed Point Theory is a fundamental area of mathematical analysis that studies conditions under which a function maps at least one point to itself. Originating from classical results such as the Banach Contraction Principle and Brouwer's Fixed Point Theorem, the theory has grown into a powerful tool used across pure and applied mathematics. It provides essential methods for proving the existence and uniqueness of solutions to differential equations, integral equations, optimization problems, and dynamic systems. The central idea of fixed points enables researchers to convert complex mathematical problems into simpler equivalent formulations. Modern developments in fixed point theory extend to metric spaces, normed linear spaces, partially ordered sets, and nonlinear mappings. The theory also plays a crucial role in computer science, economics, game theory, population models, and machine learning. This work highlights the foundational principles, important theorems, applications, and emerging research directions of fixed point theory.

Keywords: *Banach Contraction Principle, Brouwer Fixed Point Theorem, Nonlinear Mappings, Contraction Mapping, Sequence of Mappings, Families of Mappings.*



MSEC-RS-2025-MATH-004

Markov Process and Markov Chain -Its Practical Application

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Abstract

Markov chains are stochastic models that describe sequences of events where the probability of each event depends only on the last attained state, encapsulating the memoryless property known as the Markov property. This abstraction has wide applicability for forecasting and prediction tasks in fields such as finance (e.g., predicting stock price trends), industry (e.g., modeling consumer behavior), and optimization problems (e.g., decision-making using Markov Decision Processes). The models rely on transition probabilities to estimate the likelihood of moving from one state to another, ultimately aiding in future state predictions and trend identification. Markov chains offer robust, scientific approaches for modeling and analysis where outcomes are dependent solely on present status rather than past trajectory. Practical Applications: Financial Forecasting: Markov chains are applied in predicting stock price trends and portfolio management by modeling the likelihood of changes in market states over time. Consumer Behavior Prediction: Used to forecast consumer selection and transitions between products or services, benefiting areas such as marketing and economics. Consumer Behavior Prediction: Used to forecast consumer selection and transitions between products or services, benefiting areas such as marketing and economics. Optimization and Decision-Making: Markov Decision Processes (MDPs) are central in decision theory, enabling modeling and solving of complex optimization scenarios where outcomes are stochastic and depend on sequential decisions.

Keywords: *Markov chain, Markov process, Stochastic modeling, Transition probability, Time series, Forecasting, Decision process, Markov Decision Process (MDP)*



MSEC-RS-2025-PHY-001

Synthesis and optical properties of 1-Carboxy-2-Methylpropan-1-Aminium Chloride 2-Azaniumyl-3-Methylbutanoate Single Crystal

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Abstract

Slow evaporation solution growth technique was employed to grow an optical single crystal of 1-Carboxy-2-Methylpropan-1-Aminium Chloride 2-Azaniumyl-3-methylbutanoate using ethanol as a solvent. It crystallizes in a monoclinic system with space group $P2_1$ was confirmed by single crystal X-ray diffraction analysis. Optical characteristics were evaluated using UV-Vis-NIR analysis; it was observed with a cut-off wavelength at 359 nm. The Urbach energy of the grown crystal was analysed and it appears to be minimum, which indicates that the 1-Carboxy-2-Methylpropan-1-Aminium Chloride 2-Azaniumyl-3-methylbutanoate has good crystallinity and Steepness parameter, Electron-phonon interaction energy was estimated. Optical properties such as extinction coefficient, optical conductivity and refractive index were measured in terms of photon energy and wavelength. Photoluminescence analysis of the material reveals great structural perfection of grown crystal.

Keywords: *UV-Vis-NIR, Urbach energy, Steepness parameter, Electron-phonon interaction energy*



Correlation Between Surface O₃ and NO₂ Concentration at a Residential Area

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Abstract

The atmosphere of the Earth balanced the extreme events of physical and chemical changes. However, due to the interference of human activities like industrial growth, the very Earth's atmosphere which acted as a protective shield has undergone dramatic change over the past few decades. This has brought about a sense of panic and an overwhelming need to have research done on the atmosphere. Understanding the trace gases present in the atmosphere and to have measurements carried out on the trace gases has become the need of the hour as these trace gases play a very major role in the composition, chemical processes, radiative interaction, changes in the climate and pollution of the atmosphere. Ozone, the most important trace gas much debated about ever since its discovery plays a key role in deteriorating the earth's atmosphere due to its physical, chemical, human health, vegetation and climatic influences. Rapid urbanization and industrialization in the Indian region in the recent decades has brought about unprecedented growth in the emission of the atmospheric pollutants like the oxides of nitrogen, carbon monoxide, methane and non-methane hydrocarbons. These pollutants in turn also act as important precursors of atmospheric ozone. Photochemical pollution is considered as the major concern. Ozone is a major photochemical pollutant, and together with nitrogen dioxide, dominates urban air pollution.

Keywords: *GLO, NO₂, Aeroqual S500, Meteorological parameter*



Carbon Electrodes in Electrochemistry: New Materials, Hybrid Designs, and Future Perspectives

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Abstract

Electrochemistry has emerged as a cornerstone for sustainable energy conversion and storage technologies, with carbon-based electrodes occupying a central role due to their abundance, tunable structure, electrical conductivity, and chemical stability. Traditional carbon materials such as graphite, activated carbon, and carbon black have long been employed in electrochemical systems, but recent advances in nanostructuring, surface functionalization, and hybridization with metal/metal oxide or heteroatom dopants have unlocked new opportunities for enhanced performance. Novel forms of carbon, including graphene, carbon nanotubes, porous carbons, and biomass-derived carbons have demonstrated superior electrochemical properties by offering high surface area, controlled porosity, and improved electron/ion transport pathways. Beyond conventional applications in batteries and supercapacitors, carbon-based electrodes are now enabling progress in electrocatalysis, sensors, fuel cells, and electrochemical water splitting. The latest research emphasizes not only improving intrinsic activity but also tailoring electrode–electrolyte interfaces, integrating sustainable precursors, and adopting scalable fabrication methods. This review provides a comprehensive overview of recent developments in carbon-based electrodes, highlighting breakthroughs in material design, hybrid composites, and novel electrochemical applications. Finally, emerging trends and challenges- such as multi-functionality, durability, green synthesis, and the transition from laboratory-scale demonstrations to real-world deployment are discussed to guide future directions in this rapidly evolving field.

Keywords: *Electrochemistry, Carbon, Nanostructures, Hybrids, Sustainability.*



MSEC-RS-2025-CHY-001

Fabrication of Polymeric Filter for Enhanced Air Quality using High Internal Phase Emulsion Technique

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Abstract

Recently with rapid urbanization, industrialisation and modernisation air pollution is worsening and the human health hazards due to toxic pollutants have become the key research topic which is the need of the hour for the burning issue of the world community. Effective pollution control results in short terms and long term benefits for the nation as a whole. Therefore, the current work is concentrated on the fabrication of polymeric monolith filters from the Poly (styrene-co-methylmethacrylate) (PSM) and TiO₂ nanoparticles using high internal phase emulsion technique. Functional group analysis for both the polymer and nanoparticles has been studied using the FT-IR spectrum. The presence of nanoparticles within the voids and the pores of PSM/TiO₂ NPs monolith were confirmed from SEM images. The size of voids and the pores were found to be in the range of 3.2 - 6.8 μm and 2.5 - 3.1 μm respectively. High thermal stability of 350°C was noted for PSM/TiO₂ NPs, which was higher than pure PSM. The pressure drop during filtration changed from 354 to 488 Pa. The above results have proved that the filter would be a potential candidate to act as an effective air filter.

Keywords: *Air filter, HIPE, polymer monolith*



MSEC-RS-2025-CHY-002

Synthesis and Photocatalytic Property of Nanomaterials

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Abstract

Nanomaterials exhibit exceptional potential for environmental remediation and renewable – energy conversion due to their large surface area, tunable band gap energy and enhanced photocatalytic properties. This work reviews synthesis strategies and correlates morphological, structural and surface features with photocatalytic performance. The common bottom – up methods are outlined with emphasis on how crystallinity, particle size, phase composition like parameters can be controlled. Representative photocatalytic metrics like rate constant and absorbance demonstrates that carefully tuned nanomaterials can achieve rapid degradation of organic contaminants and pollutants.

Keywords: *Nanomaterial, botton-up methods, photocatalytic property, organic pollutant.*



MSEC-RS-2025-CHY-003

Synthesis, Size Characterization and Photocatalytic Studies on the Degradation of Drugs by Silver Nanoparticles

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Abstract

Metal nanoparticles show enhanced physical and chemical properties when compared to bulk materials. Particularly Silver nanoparticles (Ag Np) show wonderful catalytic, optical and spectroscopic properties. We report here synthesis, size characterization and photocatalytic behavior of Ag Np. Silver nanoparticles are investigated on the degradation of potential drugs like Isoxsuprine and Amoxicillin. In recent years it was deducted that environmental water resources have been loaded in exceeding levels by these drugs. The kinetics of degradation of these drugs in the presence of Ag Np are carried out and the results are put forth.

Keywords: *Silver nanoparticles, degradation, Isoxsuprine, Amoxicillin*



MSEC-RS-2025-CHY-004

Role of Solvents and Promoters in Enhancing Polymer Yield: A Focus on Ionic Liquid–Mediated Polymerization of Acrylamide

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Abstract

The yield and structural quality of polymers synthesized through free-radical pathways depend strongly on the choice of solvent and the presence of suitable promoters. Ionic liquids have emerged as highly efficient solvents for acrylamide polymerization due to their high polarity, thermal stability, tunable viscosity and unique charge-stabilizing environment. These properties enhance monomer solvation and facilitate superior chain propagation, resulting in improved polymer yield. The incorporation of a compatible promoter can further amplify the functional performance of ionic liquids by strengthening charge-transfer interactions, enhancing radical initiation and stabilizing reactive intermediates. This review highlights promoters that exhibit strong compatibility with ionic-liquid media. Among them, triethanolamine (TEA) and N,N-dimethylaniline (DMA) demonstrate significant potential due to their electron-donating ability and their capacity to improve solvent polarity, ionicity and charge-transfer complex formation. These promoters augment radical generation kinetics and optimize the physicochemical environment of the ionic liquid, ultimately producing polymers with higher conversion efficiency and enhanced molecular characteristics. The synergistic use of ionic liquids with tailored promoters represents a promising strategy for achieving high-yield, controlled polymerization systems.

Key words: ionic liquids, Promoters, Triethanolamine (TEA), N,N-dimethylaniline (DMA), Charge-transfer complex, Solvent effects.



MSEC-RS-2025-CHY-005

Extraction of Colchicine from the Corms of *Gloriosa Superba* using Solvent Extraction

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Abstract

Gloriosa superba or *Colchicum autumnale* is a subtropical plant whose corms are used in the treatment of inflammation, arthritis, alopecia and skin infections. Colchicine is an alkaloid present in the corms of *Colchicum autumnale* extracted by column chromatographic extraction with ethanol. It is tested for various medical studies like treatment of cancer, treatment of Gout because Colchicine is a well-known microtubule targeting agent. It dissociates into tubulin dimers. The extracted colchicine was characterized by UV and FT-IR studies.

Keywords: *Gloriosa superba*, *Colchicine*, *Arthritis*



Beyond Academics: A Study on Enhancing Confidence, Communication, and Soft Skills for Student Employability in the Post-Covid Era

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Abstract

The Covid-19 pandemic has fundamentally altered the educational landscape, leaving lasting effects on students' learning habits, social interactions, and employability skills. Though academic knowledge remains essential, employers increasingly seek graduates who demonstrate confidence, effective communication, adaptability, and other soft skills. Yet, many students continue to struggle in these areas, resulting in challenges during placement drives. These gaps are not merely technical but behavioral and cognitive, shaped by reduced face-to-face interactions, prolonged online learning, and the pervasive influence of popular media, which has eroded reading habits, attention spans, and meaningful social engagement. This literature-based study aims to explore the factors contributing to deficiencies in confidence, communication, and soft skills among students in the post-Covid era. By reviewing peer-reviewed research, institutional reports, surveys, and empirical studies, the research seeks to identify patterns in student behavior, challenges in employability, and areas where educational interventions are necessary. The study highlights the importance of understanding post-pandemic behavioral and cognitive changes in students to support their holistic development. It aims to provide insights for educators, administrators, and policymakers to help students navigate the transition from academics to professional environments effectively, emphasizing the need to go beyond academics in preparing students for the workforce.

Keywords: *Employability, Skill gaps, Post-covid era, cognitive, Academic behaviour*



MSEC-RS-2025-ENG-002

A Glimpse of Cultural Depletion in the Regional Novels with Special Reference to M.T. Vasudevan Nair's Naalukettu and Habeechi's Tharar

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Abstract

India is one the most linguistically diverse countries in the world. The Constitution of India recognizes 22 official regional languages in India with a developed script and literary tradition. Linguistic diversity adds to the richness of Indian culture and every region of India has a highly developed literary tradition with excellent writers of national, and a few of international fame. The government of India has committed itself to the preservation and promotion of every regional language and literature in India through institutions such as the Sahitya Akademi. For generations, India has had a prevailing tradition of the joint family system. It is a system under which extended members of a family – parents, children, the children's spouses and their offspring, etc. – live together. Usually, the oldest male member is the head in the joint Indian family system. He mostly makes all important decisions and rules, and other family members are likely to abide by them. But gradually such a tradition has transformed into modernity where the human beings are running behind money and power. Such a change will finally end up in the world of virtual reality where emotions are hardly seen and persons with emotions can be kept as a specimen in laboratories. This paper is likely to discuss the cultural diversity of two regional literary novels, originally written in Malayalam, Naalukettu (The House around the Courtyard) by M. T. Vasudevan Nair and Asamayam (Tharar, In the Shadows of Time) by Habeeb C. Rahman or Habeechi, which highlights the two extended cultures that a region faced and yet to face in future.

Keywords: *Regional languages, tradition, modernity, virtual reality, cultural diversity*



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